

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Week 1

Why Physics is Important for Computer Science?

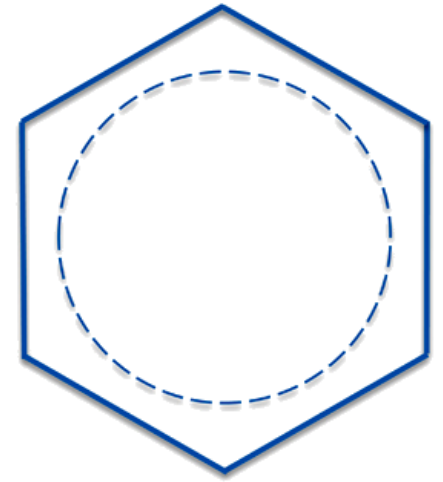
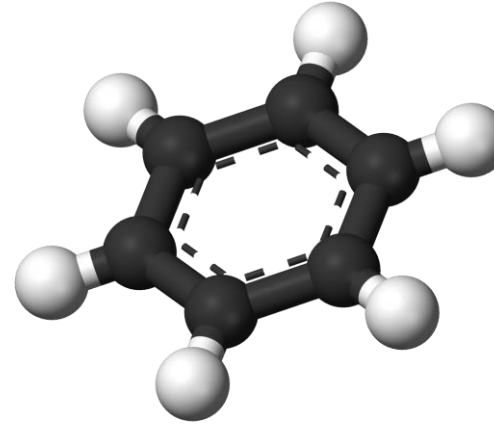
Muhammad Daniyal Chundrigar

سَنُرِيهِمْ ءَايَاتِنَا فِي الْآفَاقِ وَفِي أَنْفُسِهِمْ حَتَّىٰ يَتَبَيَّنَ لَهُمْ أَنَّهَ الْحَقُّ ۚ أَوَلَمْ يَكْفِ بِرَبِّكَ أَنَّهُ عَلَىٰ كُلِّ شَيْءٍ شَهِيدٌ ۝٥٣

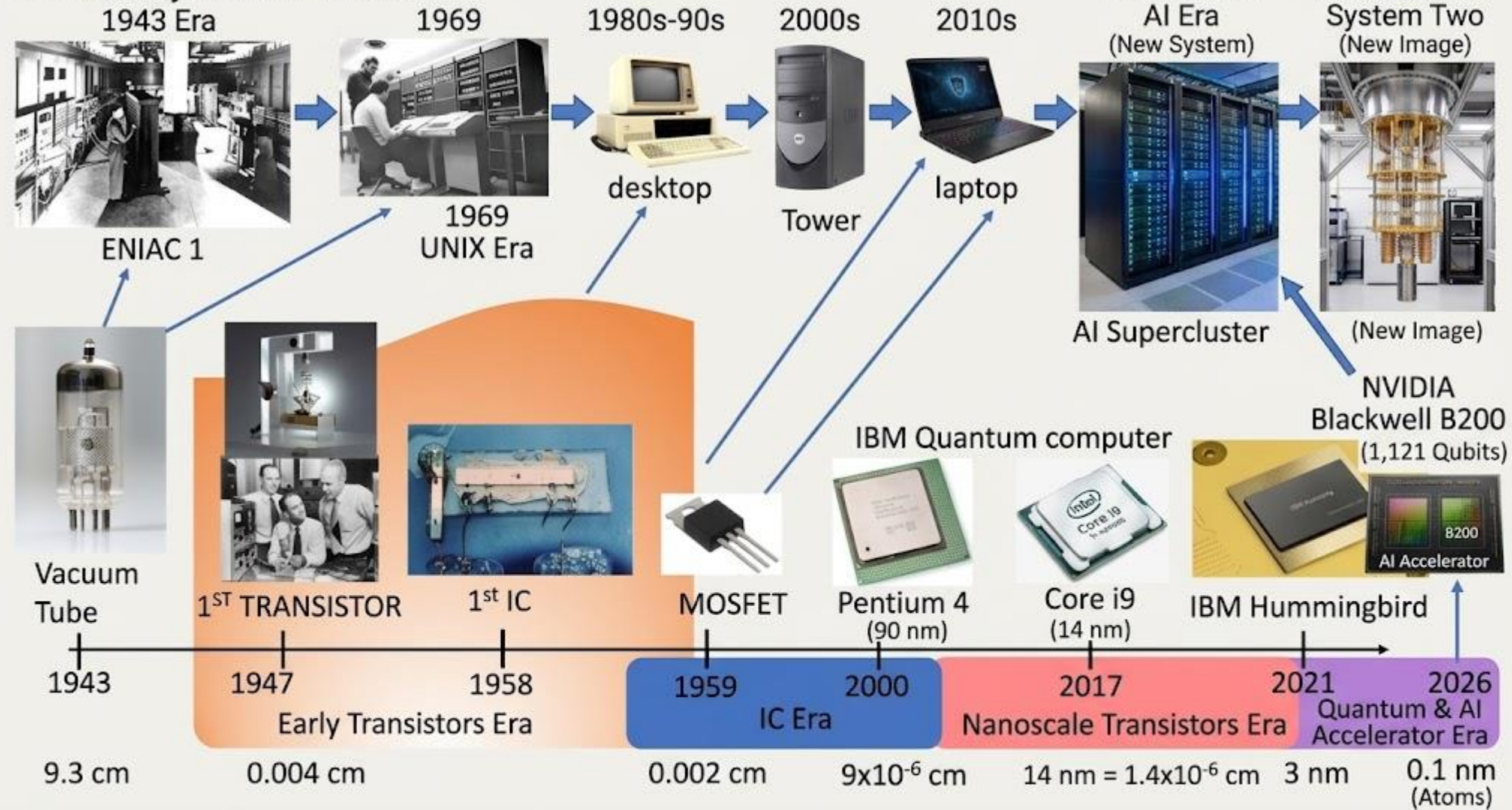
Soon shall We show them Our Signs on the horizons and in their own beings until it becomes clear to them that it is the Truth. Is it not enough that your Lord is a witness over everything?

عنقریب ہم انہیں دکھائیں گے اپنی نشانیاں آفاق میں بھی اور ان کی اپنی جانوں کے اندر بھی یہاں تک کہ ان پر واضح ہو جائے گا کہ یہ (قرآن) حق ہے کیا یہ بات کافی نہیں ہے کہ آپ کا رب ہر چیز پر گواہ ہے !

(Surah Fussilat, 41:53)

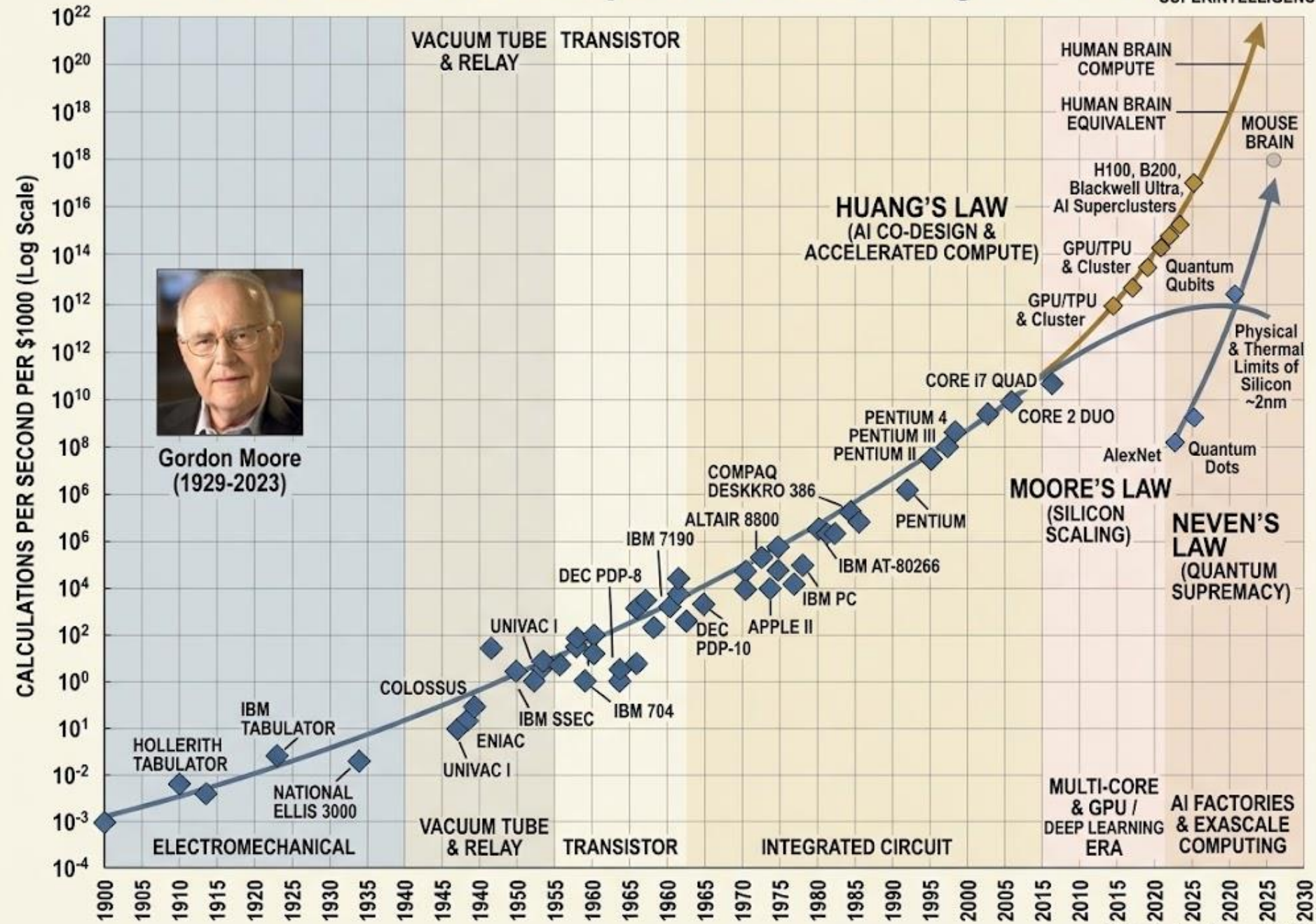


Hardware Systems Evolution

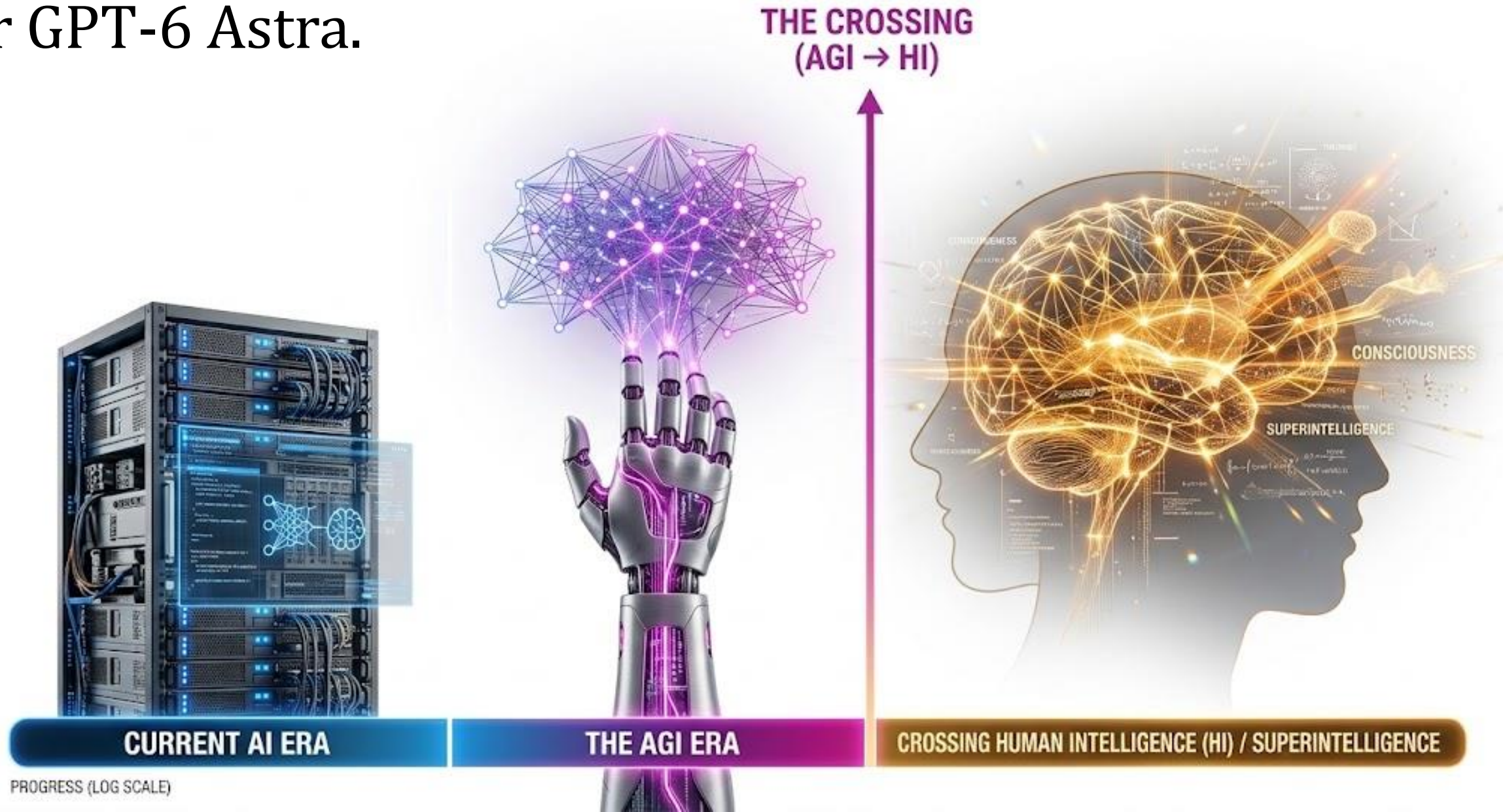


The Evolution of Compute & Accelerating Returns

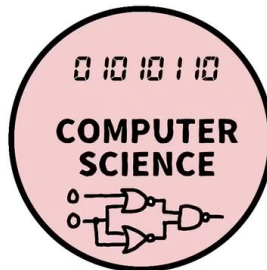
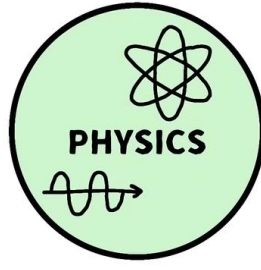
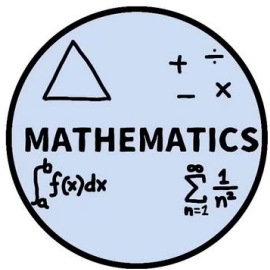
AI SWARMS /
SUPERINTELLIGENCE

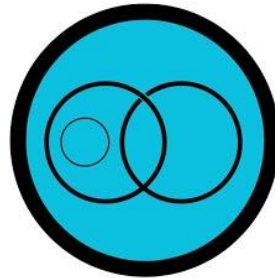
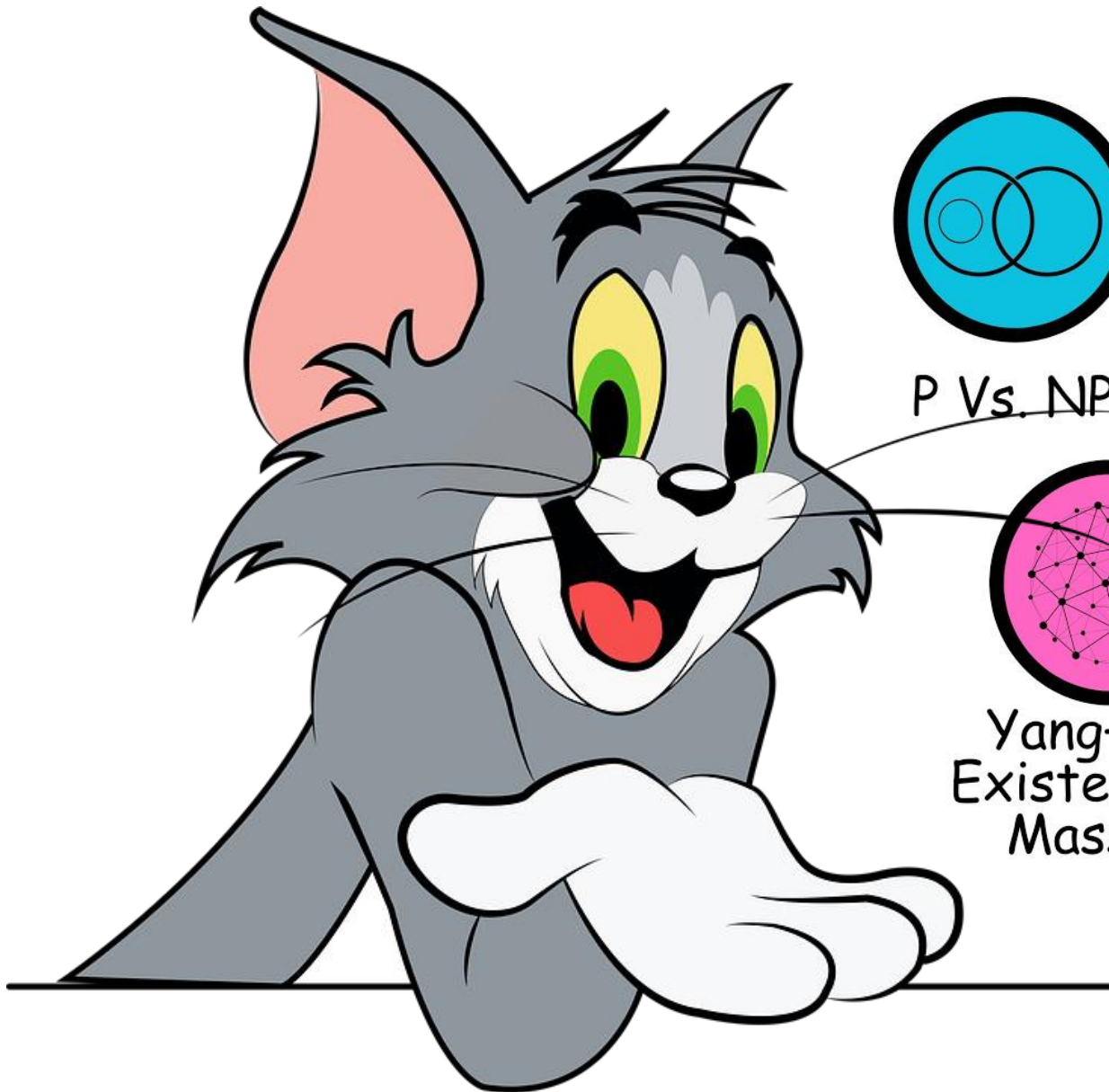


Now we have hit the AGI era After GPT-6 Astra.



The Millennium Prize Problems

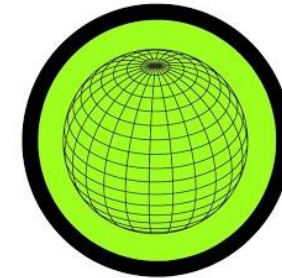




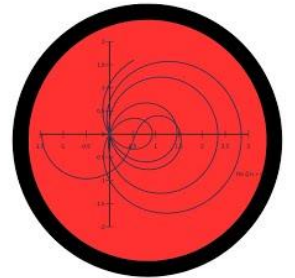
P Vs. NP



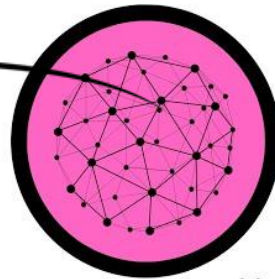
Hodge
Conjecture



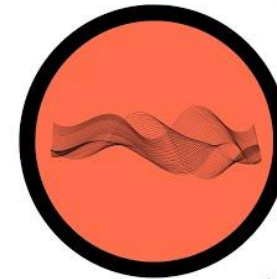
Poincaré
Conjecture



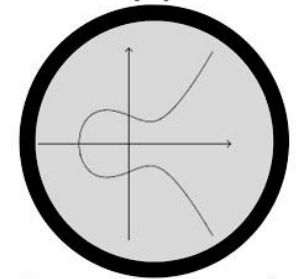
Riemann
Hypothesis



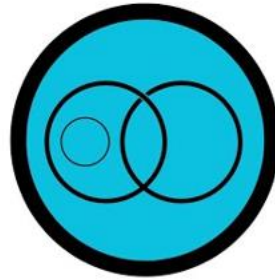
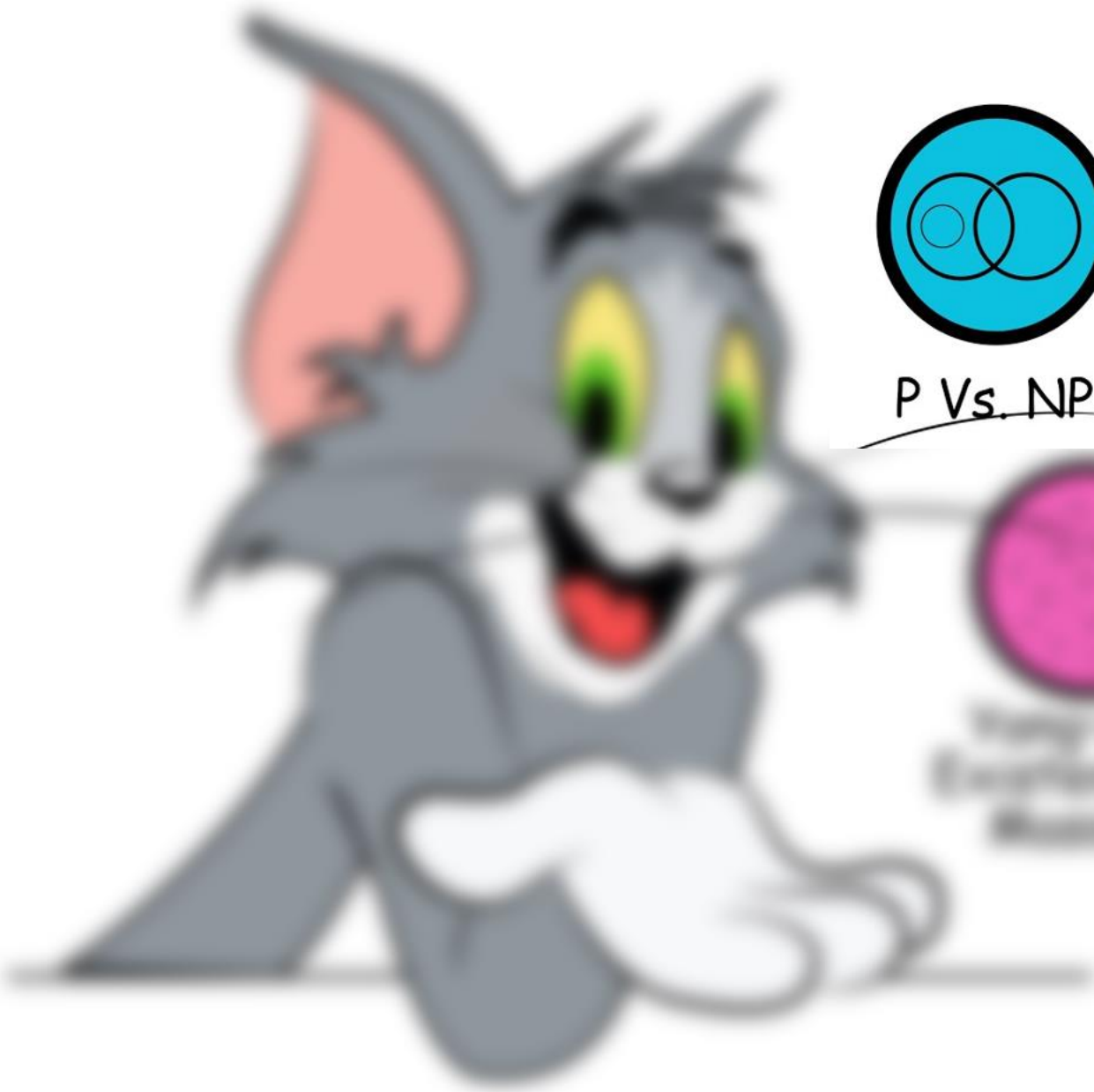
Yang-Mills
Existence and
Mass Gap



Navier-Stokes
Existence and
Smoothness



Birch and
Swinnerton-Dyer
Conjecture



P Vs. NP



Traveling Salesman Problem



Traveling Salesman Problem



Traveling Salesman Problem



Traveling Salesman Problem

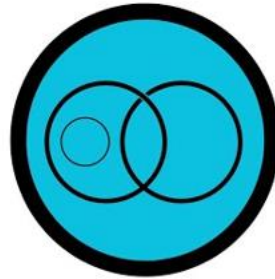


Traveling Salesman Problem



Traveling Salesman Problem

Finding a solution is
fundamentally harder than
verifying one



P Vs. NP



Traveling Salesman Problem



Traveling Salesman Problem



Traveling Salesman Problem



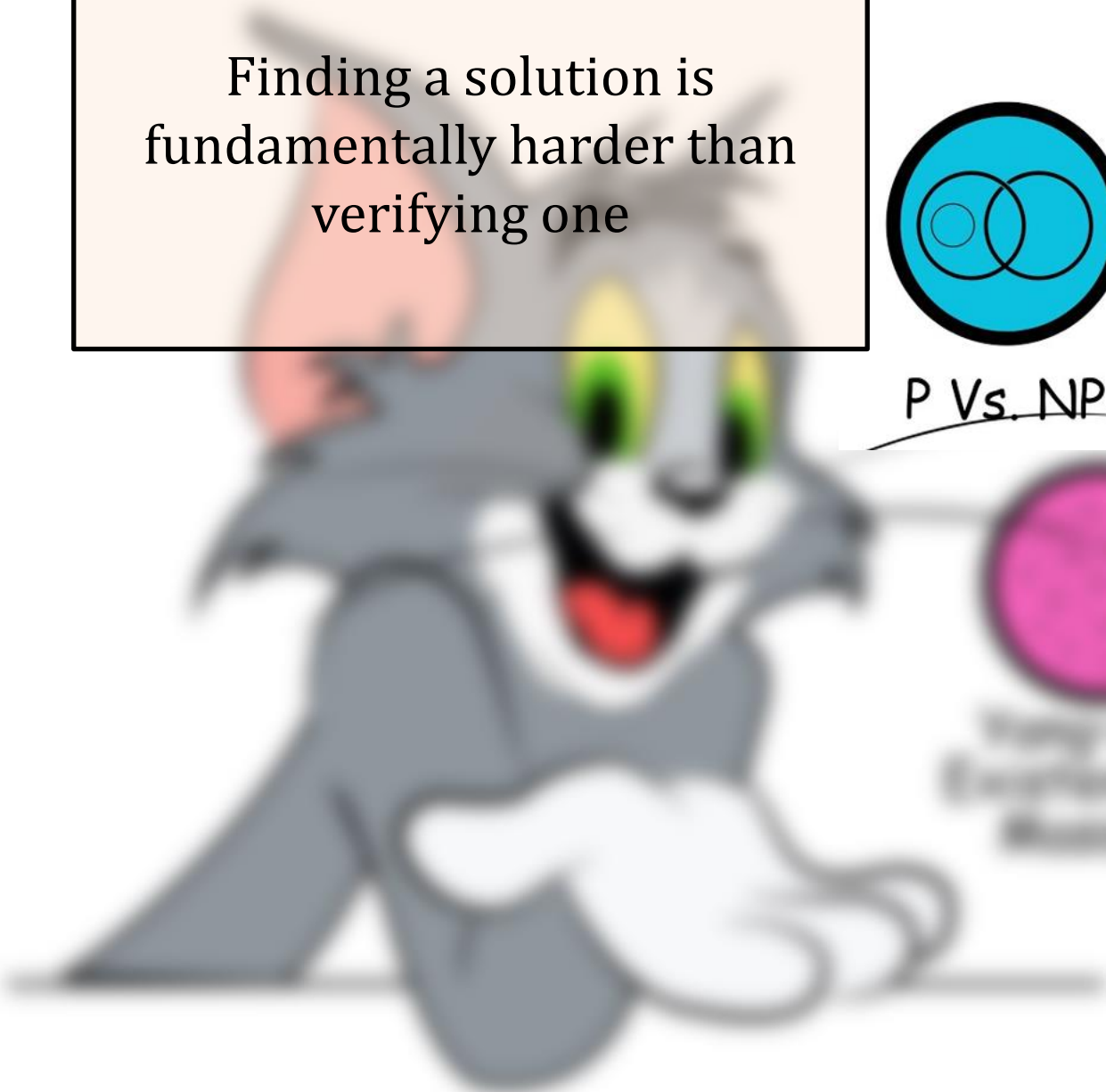
Traveling Salesman Problem



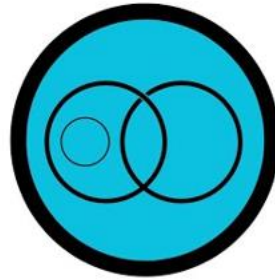
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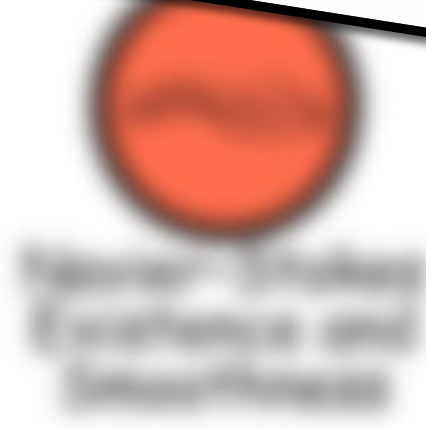
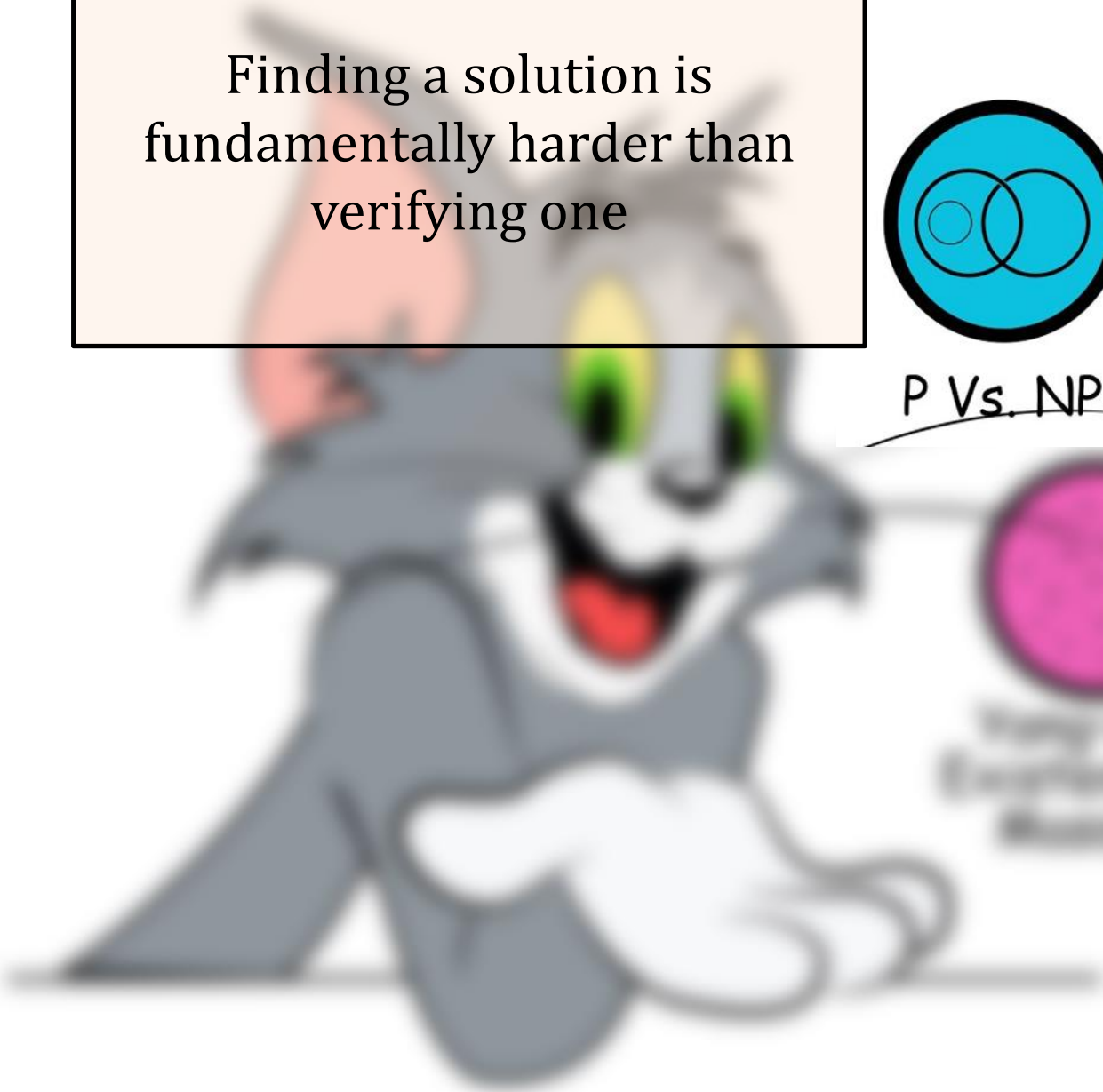
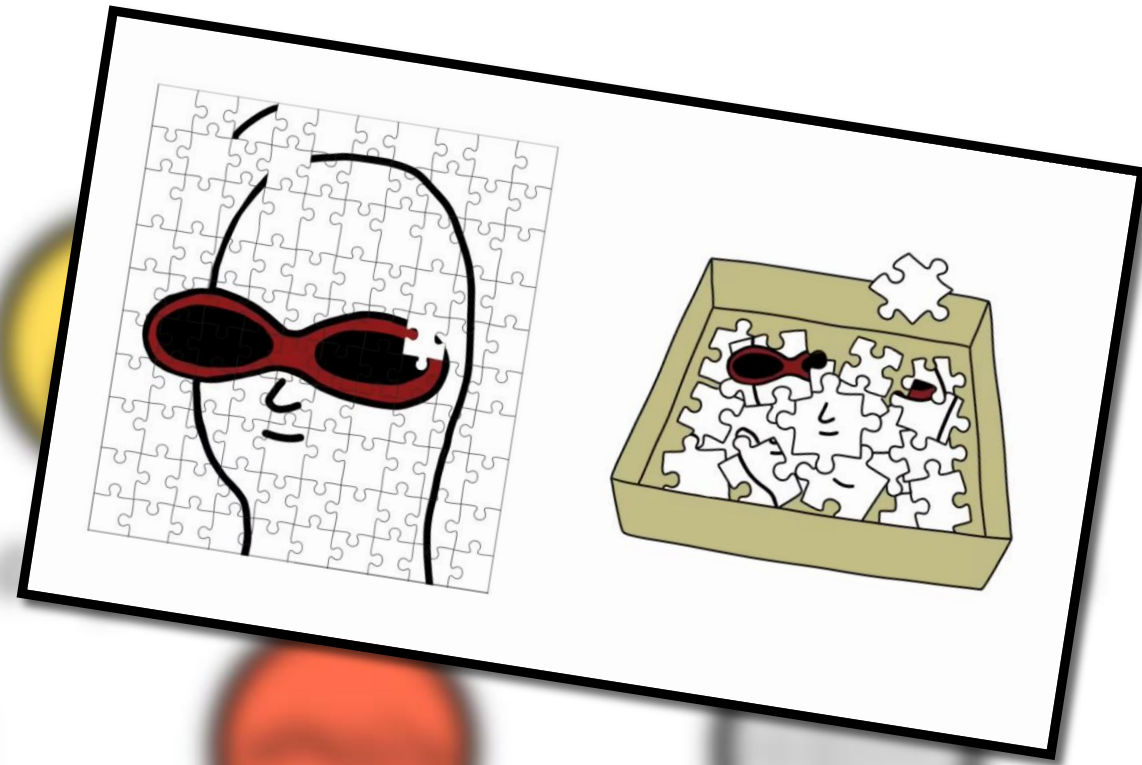
Traveling Salesman Problem



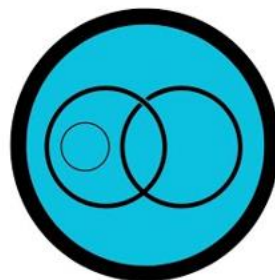
Finding a solution is fundamentally harder than verifying one



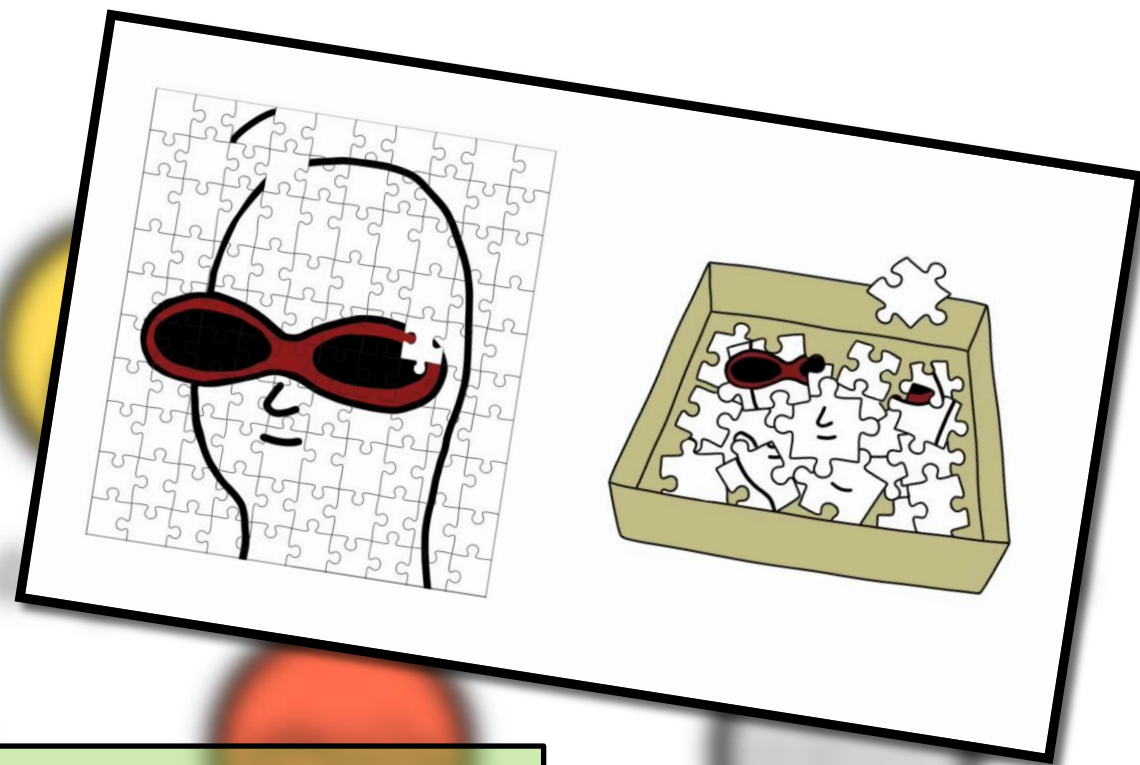
P Vs. NP



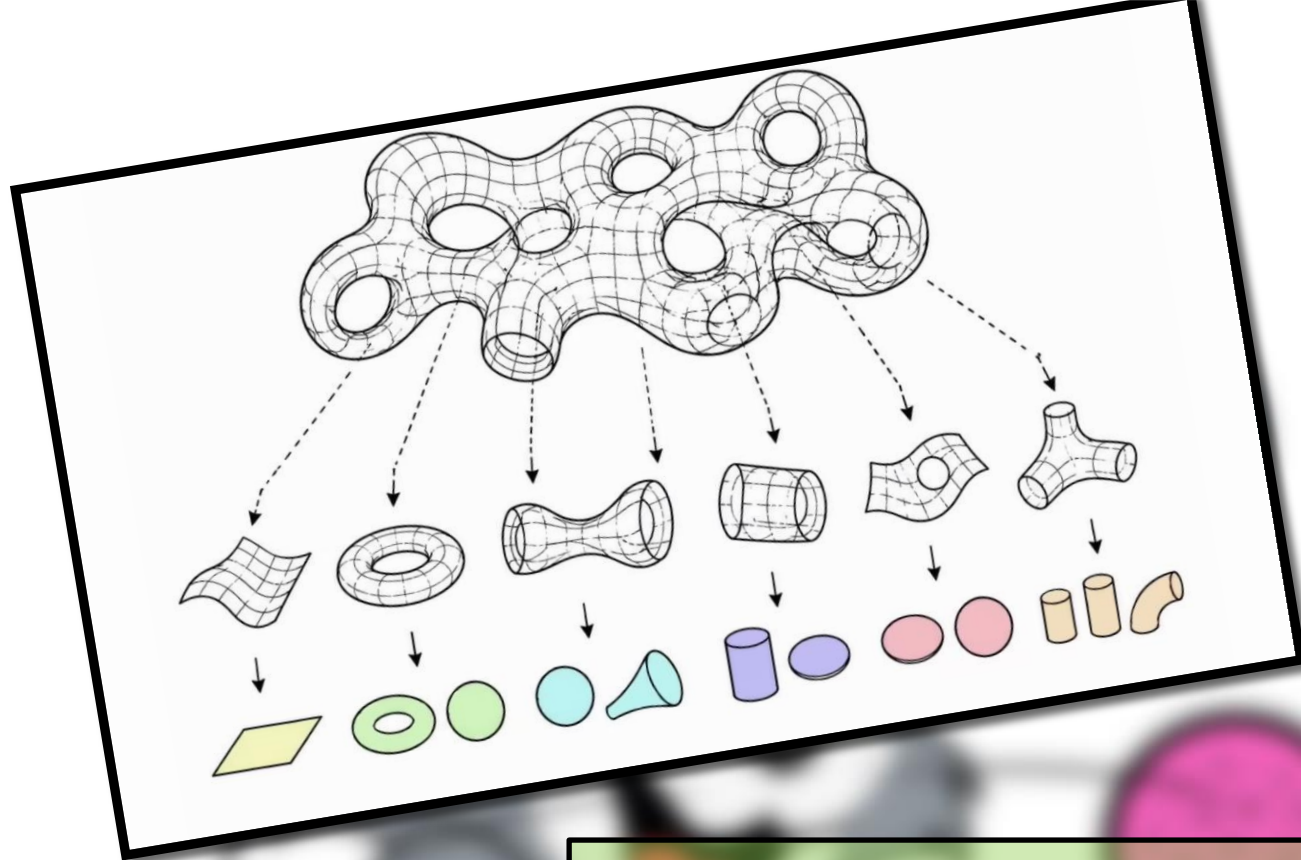
Finding a solution is
fundamentally harder than
verifying one



P Vs. NP



The Big Question is: The problem is easy
to check, is it also easy to solve ?



Hodge
Conjecture

The conjecture suggests these complex structure can always be broken down into simpler geometric building blocks

The idea is obvious but proving it has defeated mathematicians for decades



Riemann Hypothesis



Navier-Stokes
Existence and
Smoothness



Poincaré
Conjecture



Birch and
Swinnerton-Dyer
Conjecture



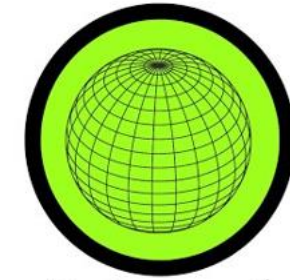
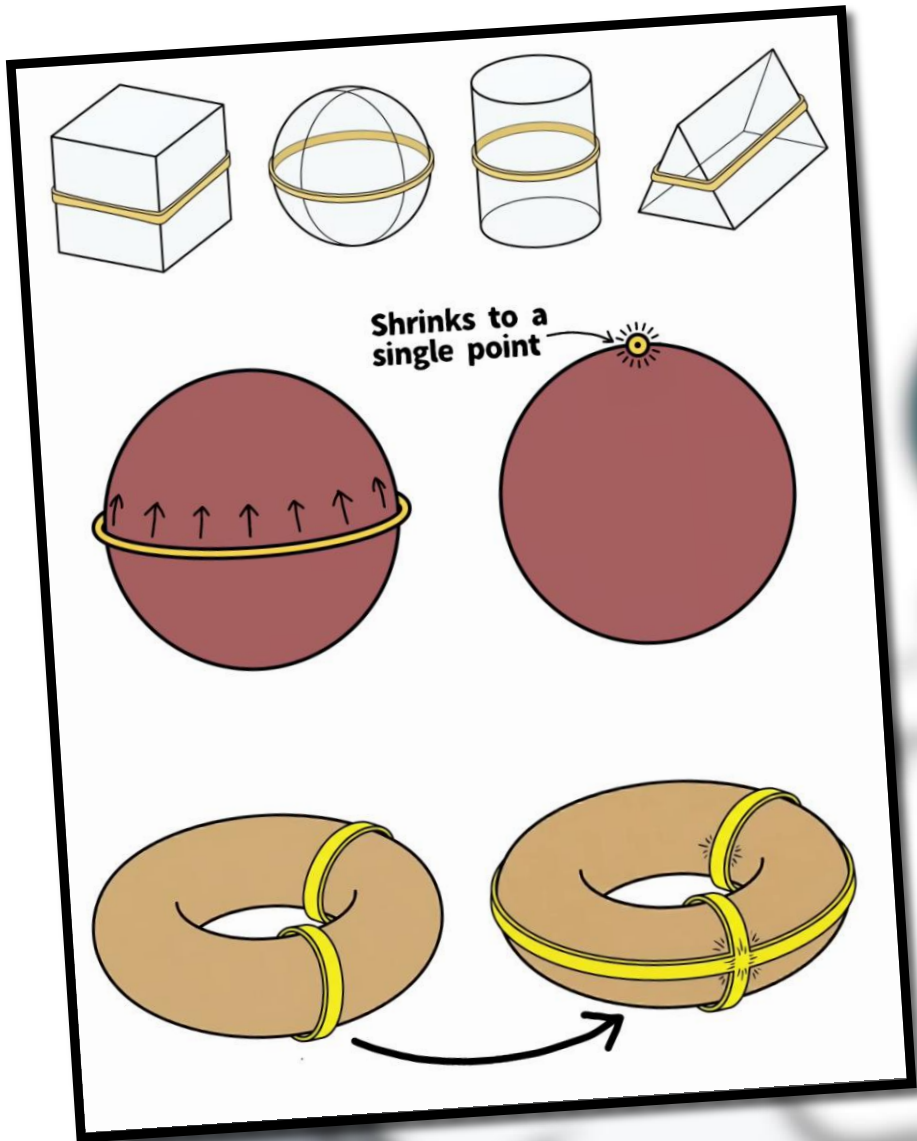
Hodge Conjecture



P vs NP
Problem



Millennium Prize
Problems



Poincaré
Conjecture



Poincaré
Conjecture



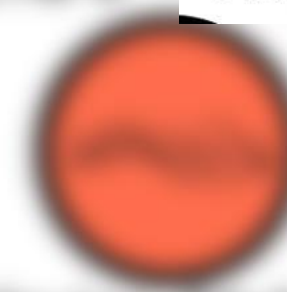
Poincaré
Conjecture



Poincaré
Conjecture



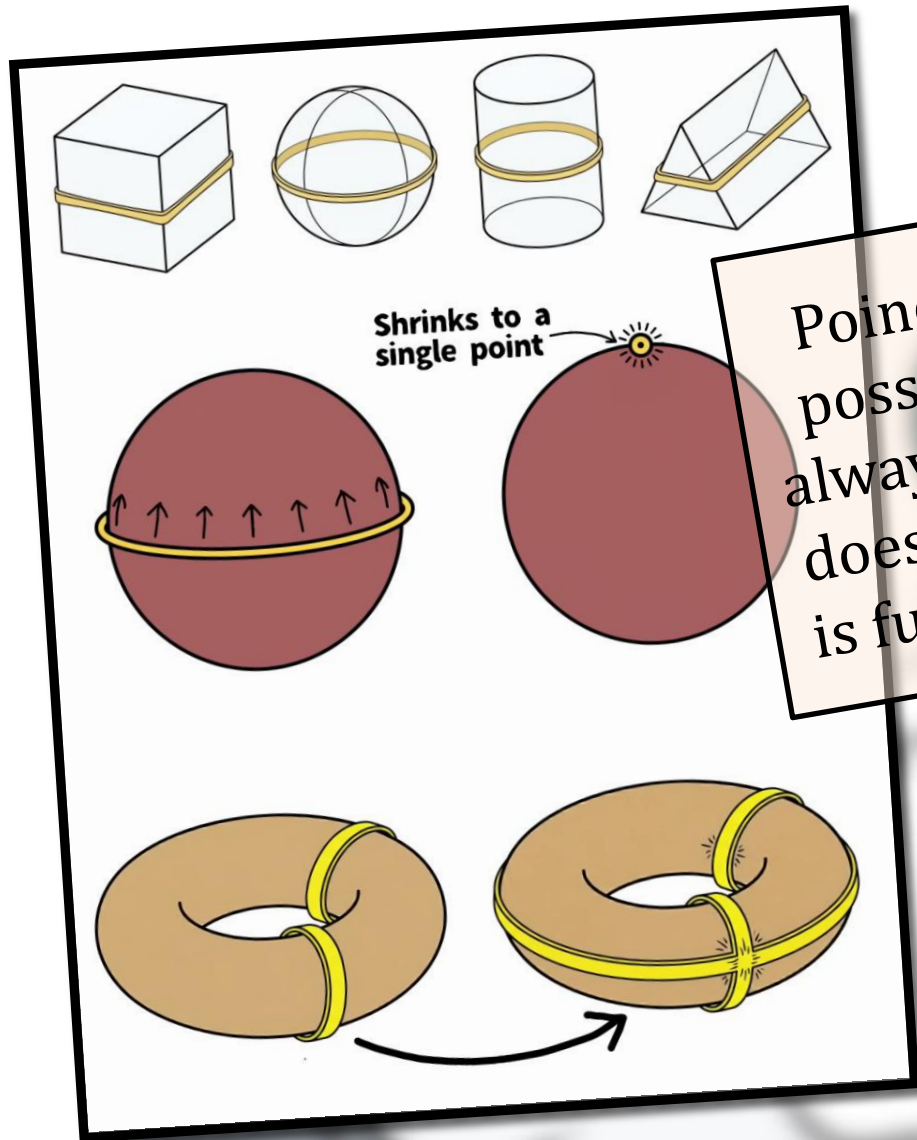
Poincaré
Conjecture



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Conjecture



Poincaré
Conjecture



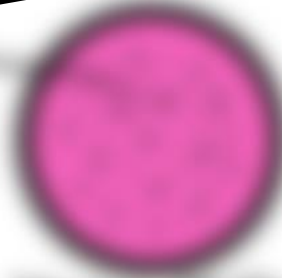
Poincare wondered, if every possible loop on a shape can always shrink down to a point, does that guarantee the shape is fundamentally sphere like??



Poincaré Conjecture



Not a sphere



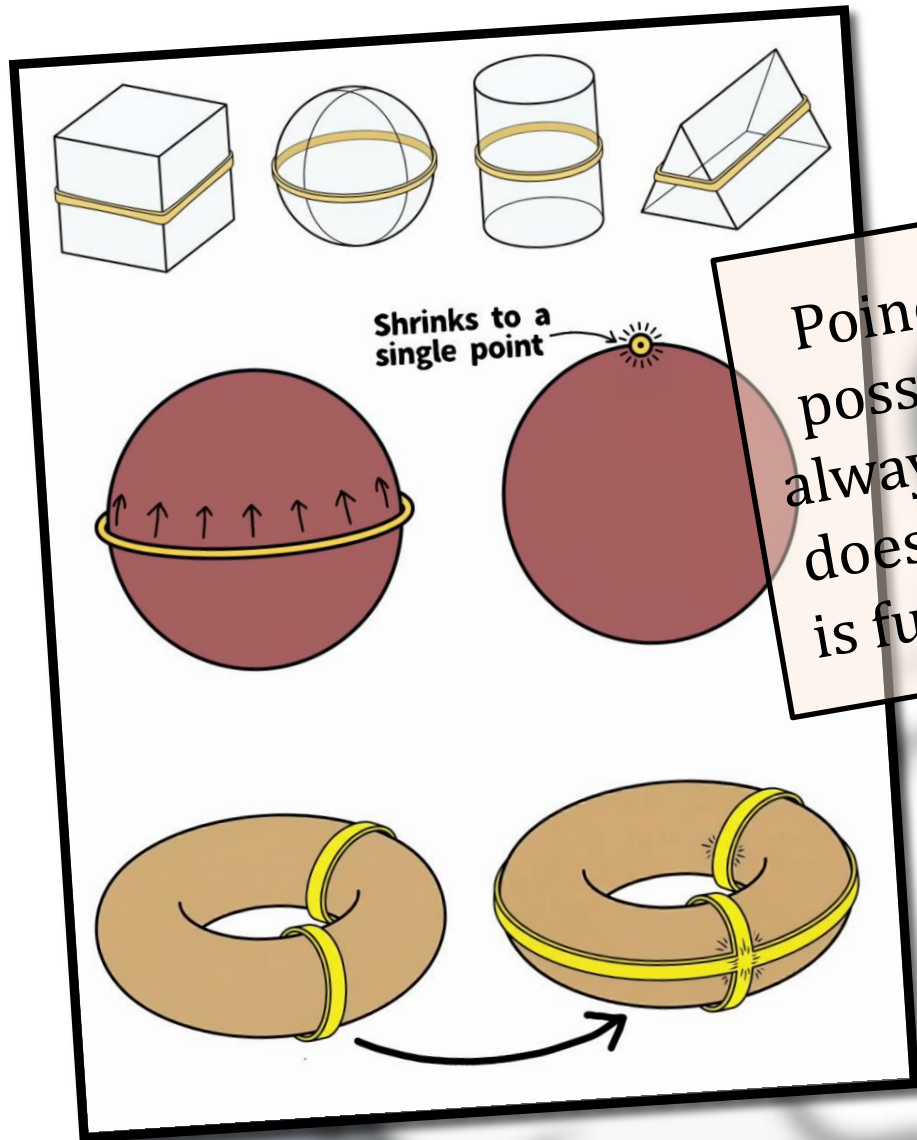
Not a sphere



Not a sphere



Not a sphere



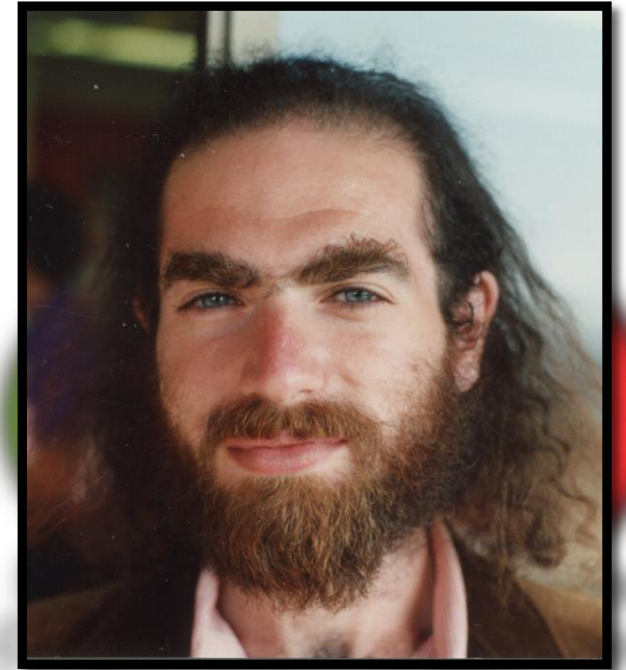
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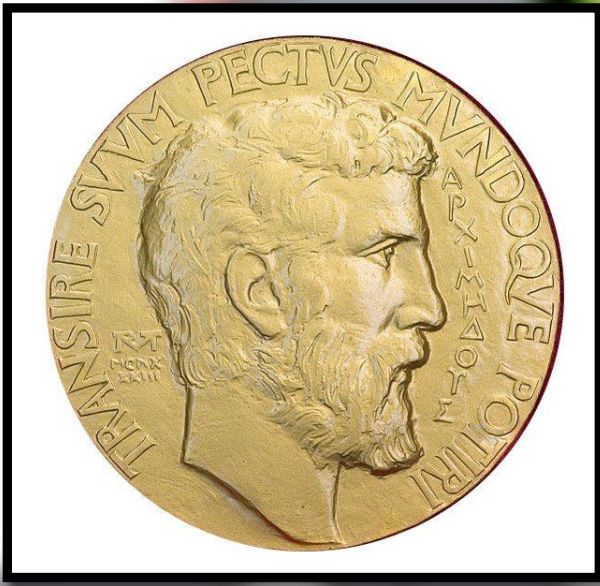
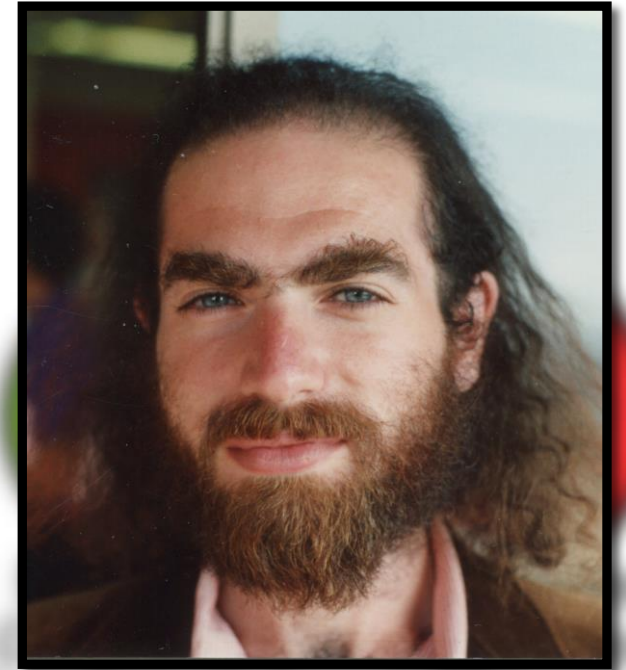
Poincaré Conjecture

In 2002, a Russian Mathematician name Grigori Perelman quietly uploaded a proof online. Years Later, the proof was confirmed to be correct.

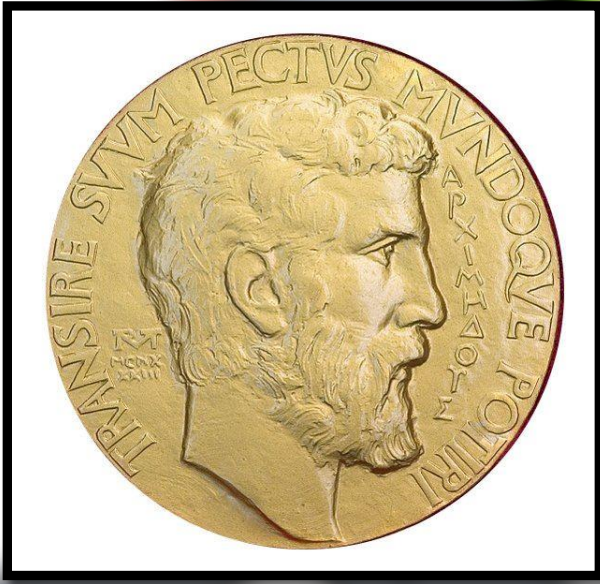
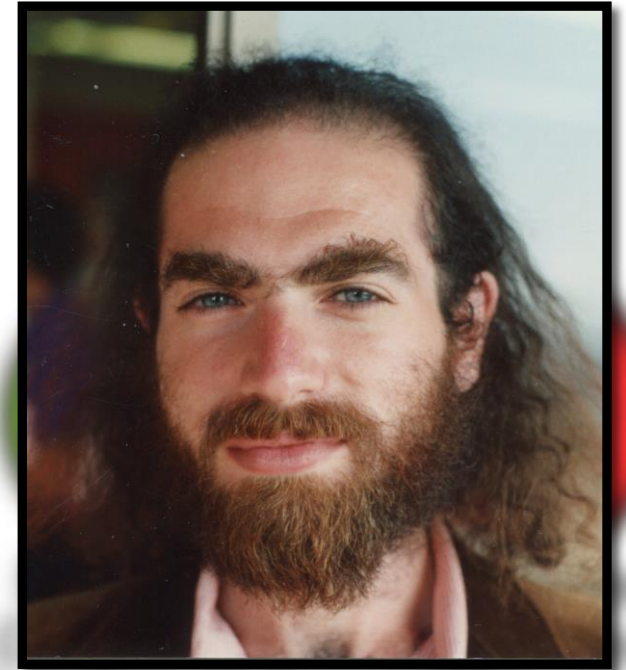
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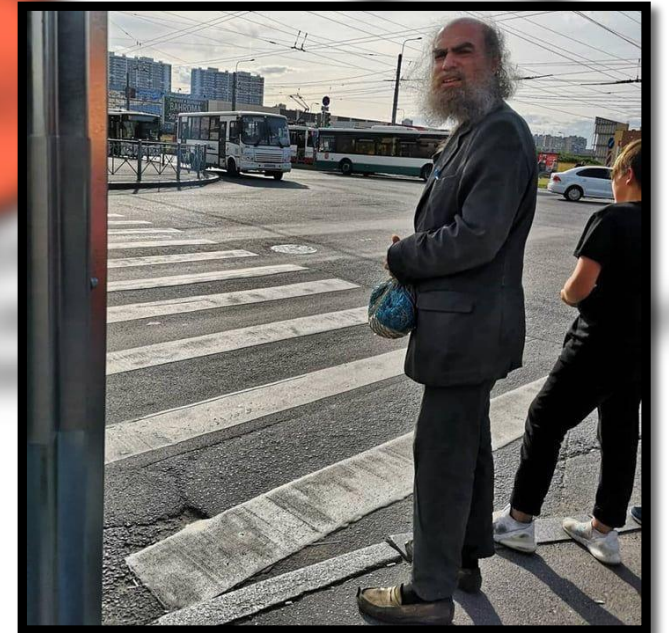
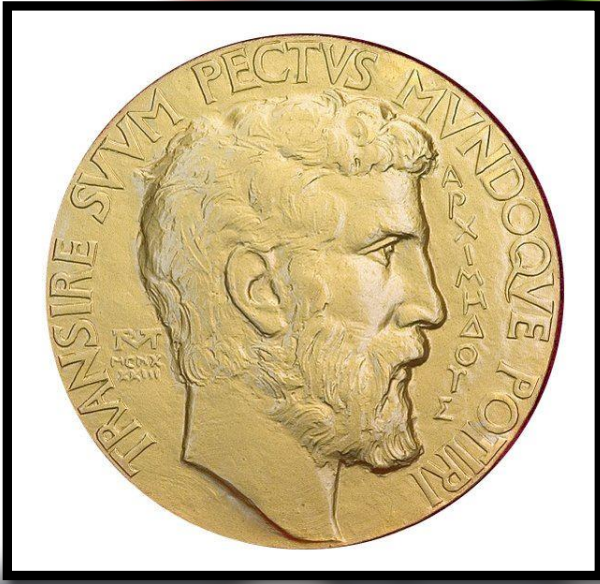
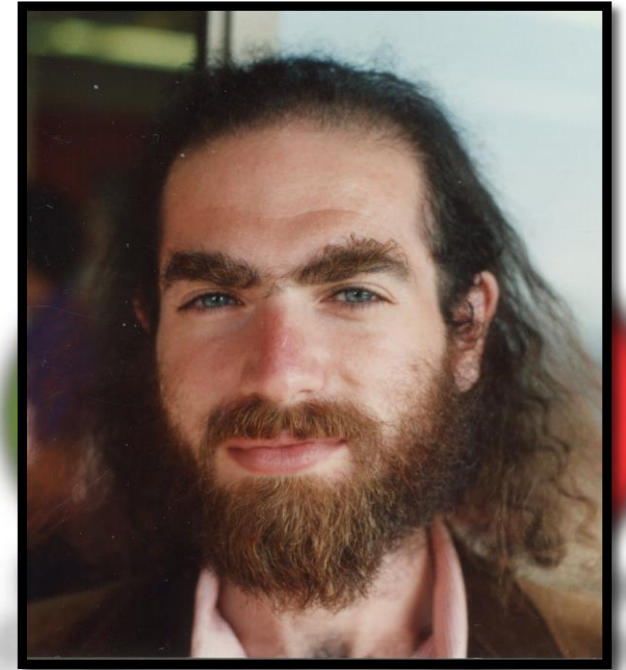
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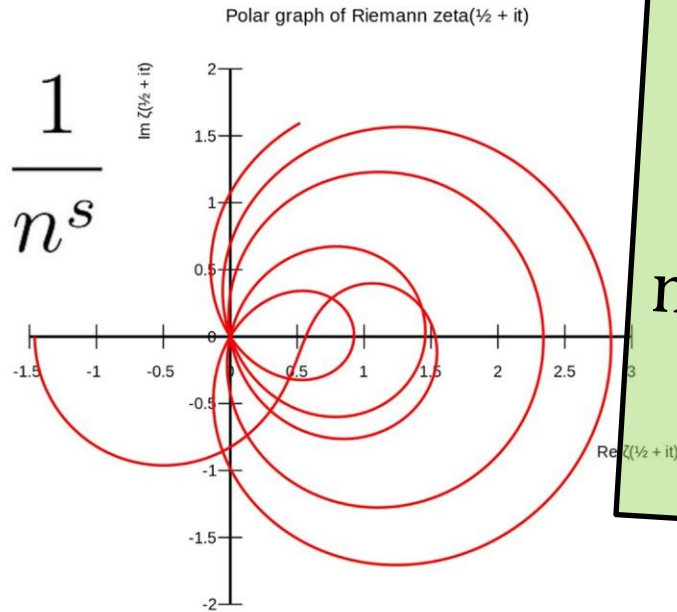


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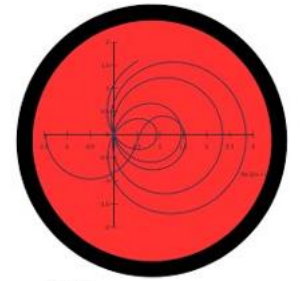


Mathematicians search pattern inside prime numbers for centuries. In 1859, Riemann discovered a strange connection between prime numbers and a mathematical object called "Zeta Function"

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

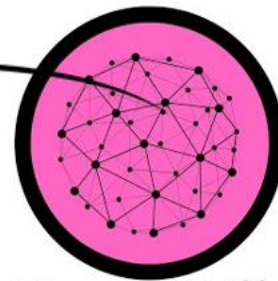
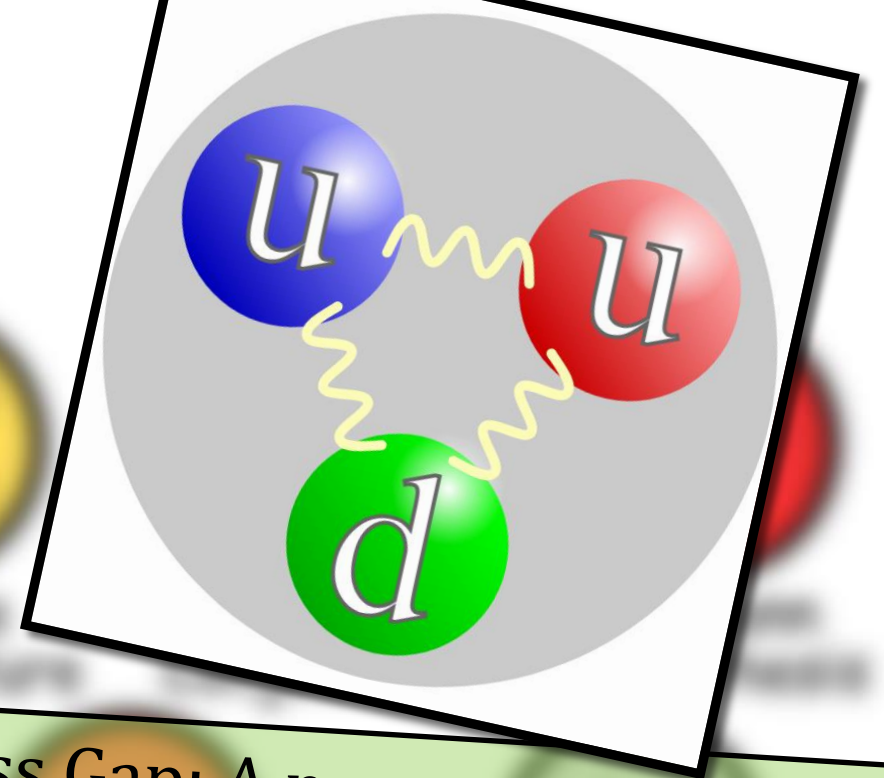


Computer check trillions of cases. Every single one matches the prediction but in mathematics example are never enough, a proof still doesn't exist.



Riemann Hypothesis

The mathematics describing this strong force is called Yang-Mills Existence.

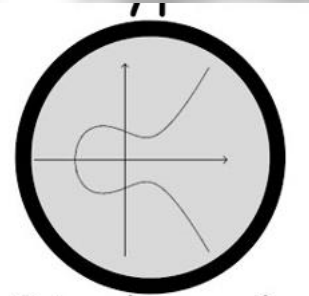
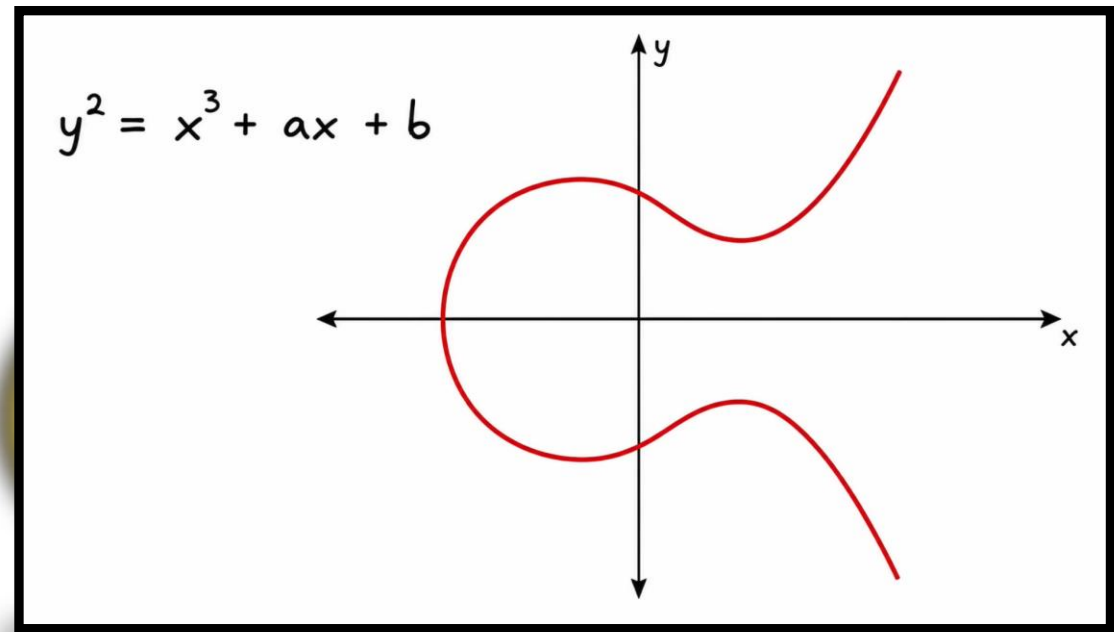


Yang-Mills
Existence and
Mass Gap

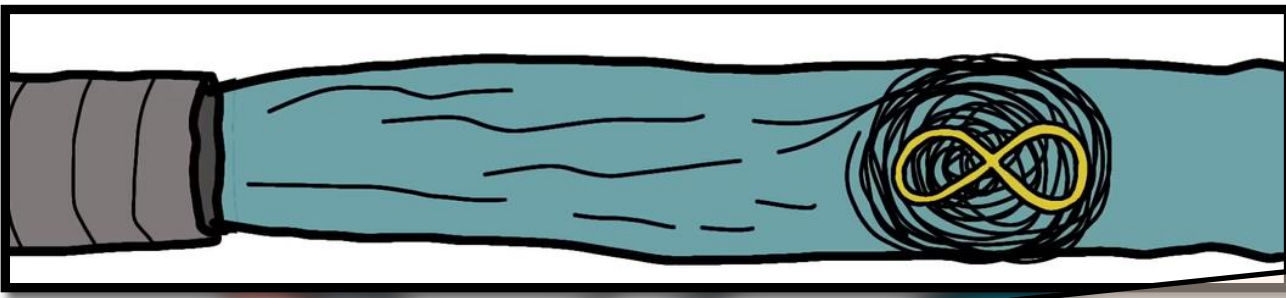
Mass Gap: A property explaining why the strong force only act across tiny distances. Physics observes it constantly. Mathematician still can't prove it why it happens.

How many rational solution does the curve
like this have?
Some say zero and some say infinite

Two mathematicians Birch and
Peter Swinnerton-Dyer
proposed that the answer could
be predicted by a some another
function connect to the curve.
But nobody proved it in general.



Birch and
Swinnerton-Dyer
Conjecture



This problem is about fluid flow. Water through pipes, air around airplanes, storms in the atmosphere.

Continuity Equation

$$\nabla \cdot \vec{V} = 0$$

Momentum Equations

$$\rho \frac{D\vec{V}}{Dt} = -\nabla p + \rho \vec{g} + \mu \nabla^2 \vec{V}$$

Total derivative

Pressure gradient

Body force term

Diffusion term

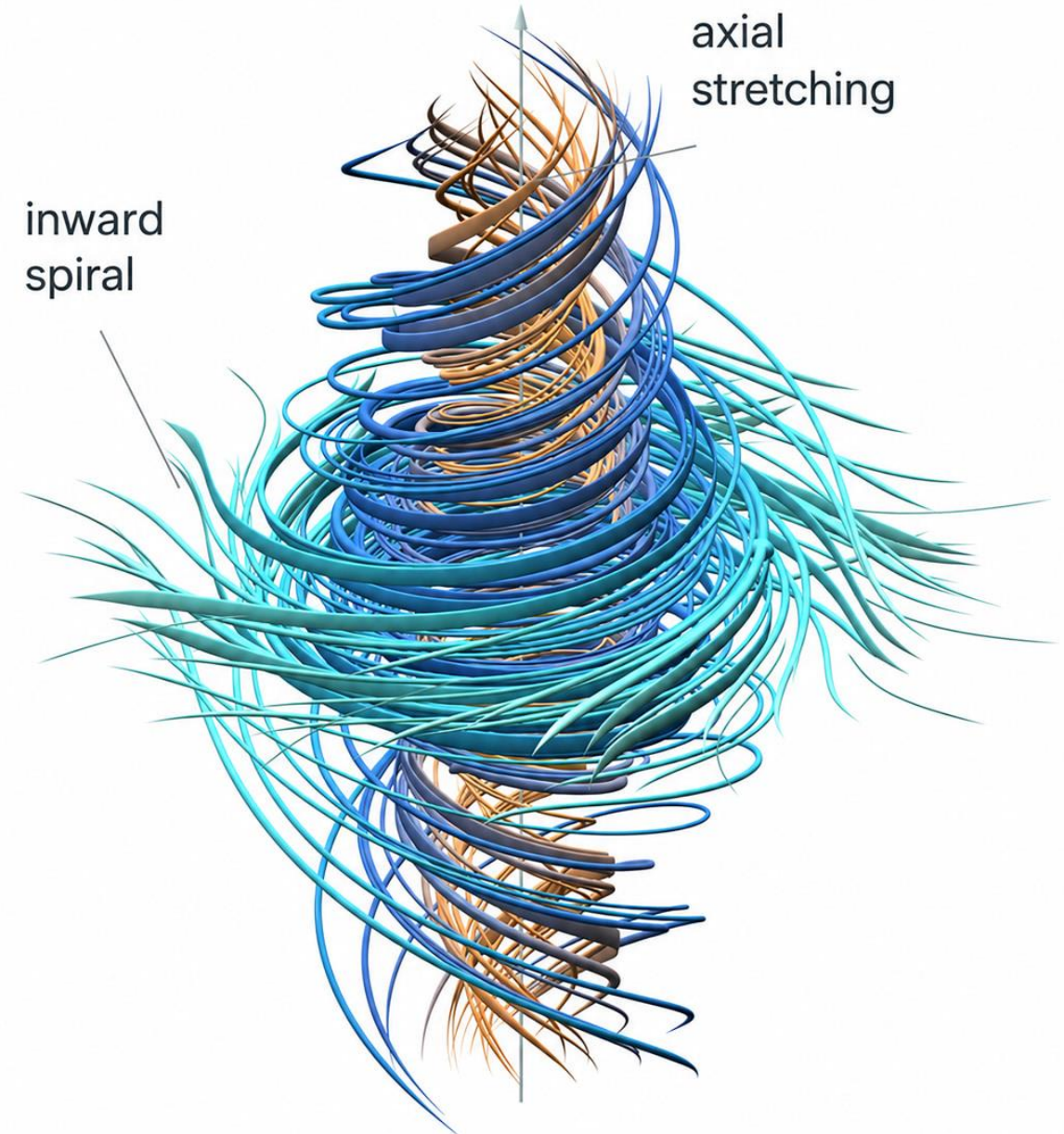
Nobody can prove the equations always behave properly. Under extreme condition the mathematics suggest fluid could become infinitely chaotic at a single point. Something that physically should not happen



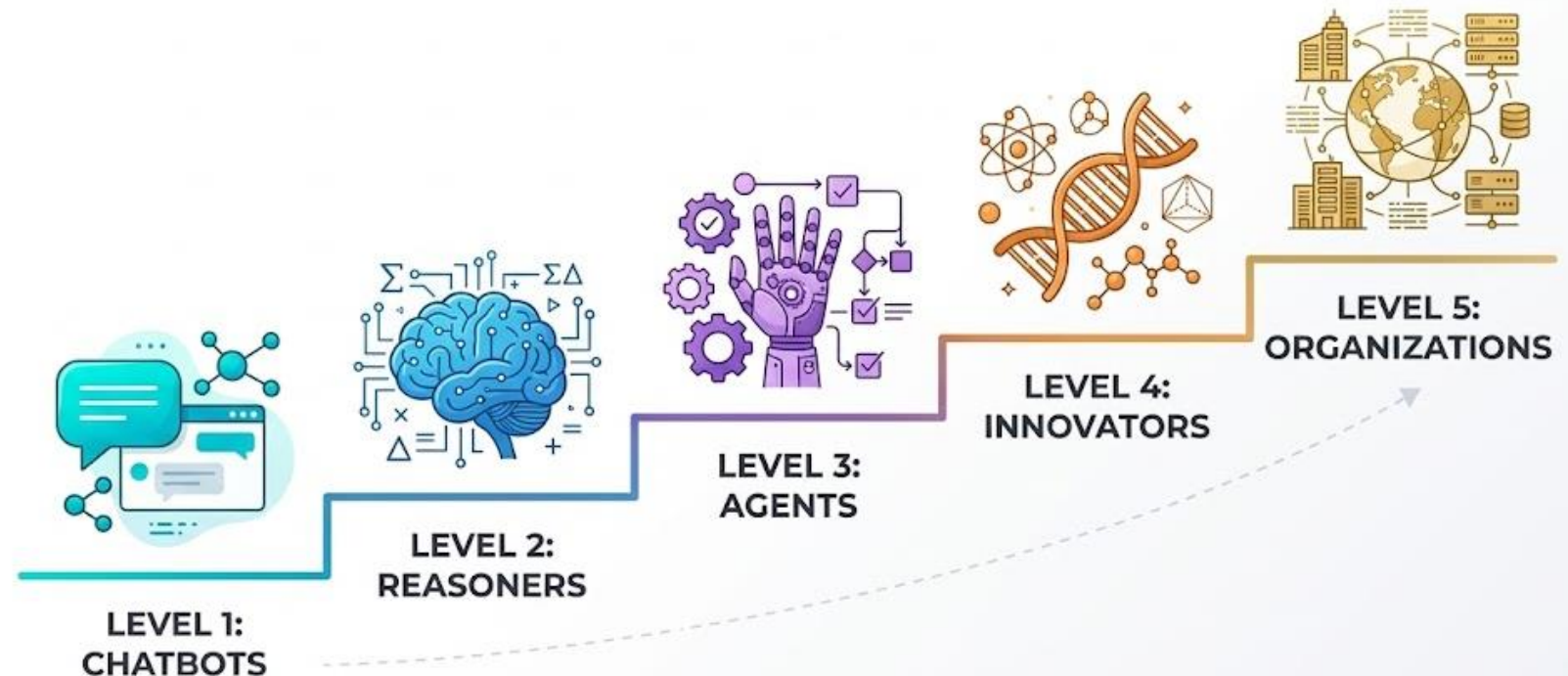
Navier-Stokes
Existence and
Smoothness

Navier-Stokes Problem has been Solved by GPTA-6 Astra Recently.

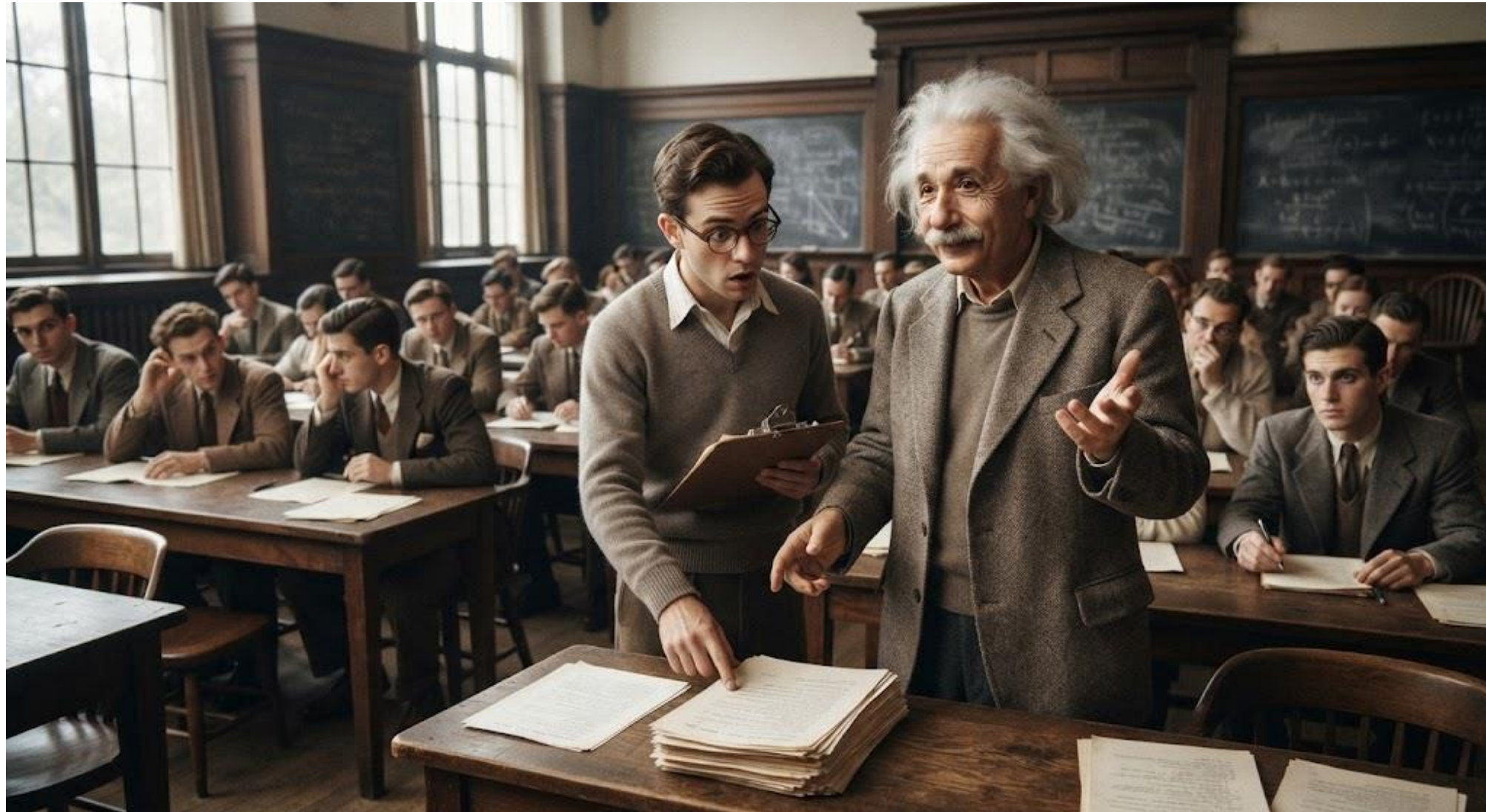
OpenAI claimed on 8 September 2026



- Level 1: Chatbots ---> Conversational AI with natural language fluency
- Level 2: Reasoners ---> Human-level problem-solving in complex domains
- Level 3: Agents ---> Autonomous systems executing multi-step workflows
- Level 4: Innovators ---> AI that invents new science, math, and technologies
- Level 5: Organizations ---> Autonomous AI entities running full enterprises



We are living in a world where the questions might be the same, but the answers have changed.





COURSE ORIENTATION & FUNDAMENTALS

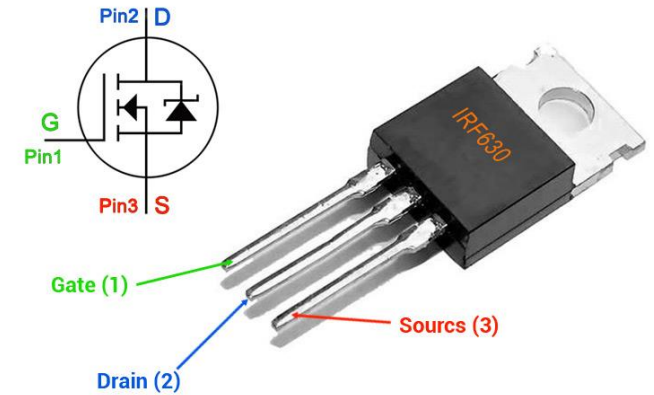
Why Physics is Important for Computer Science?

Bridging Silicon Hardware, Circuit Limits, Signal Transmission, Artificial Intelligence and Quantum Technologies

The Physical Foundation of Digital Logic

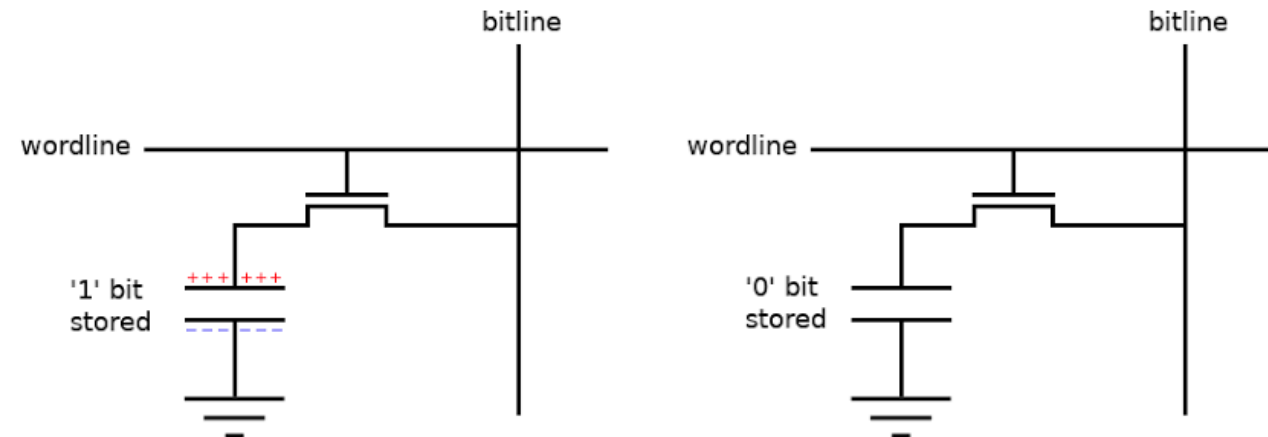
Semiconductor Physics

MOSFET transistors use semiconductor band gap dynamics to switch binary states.

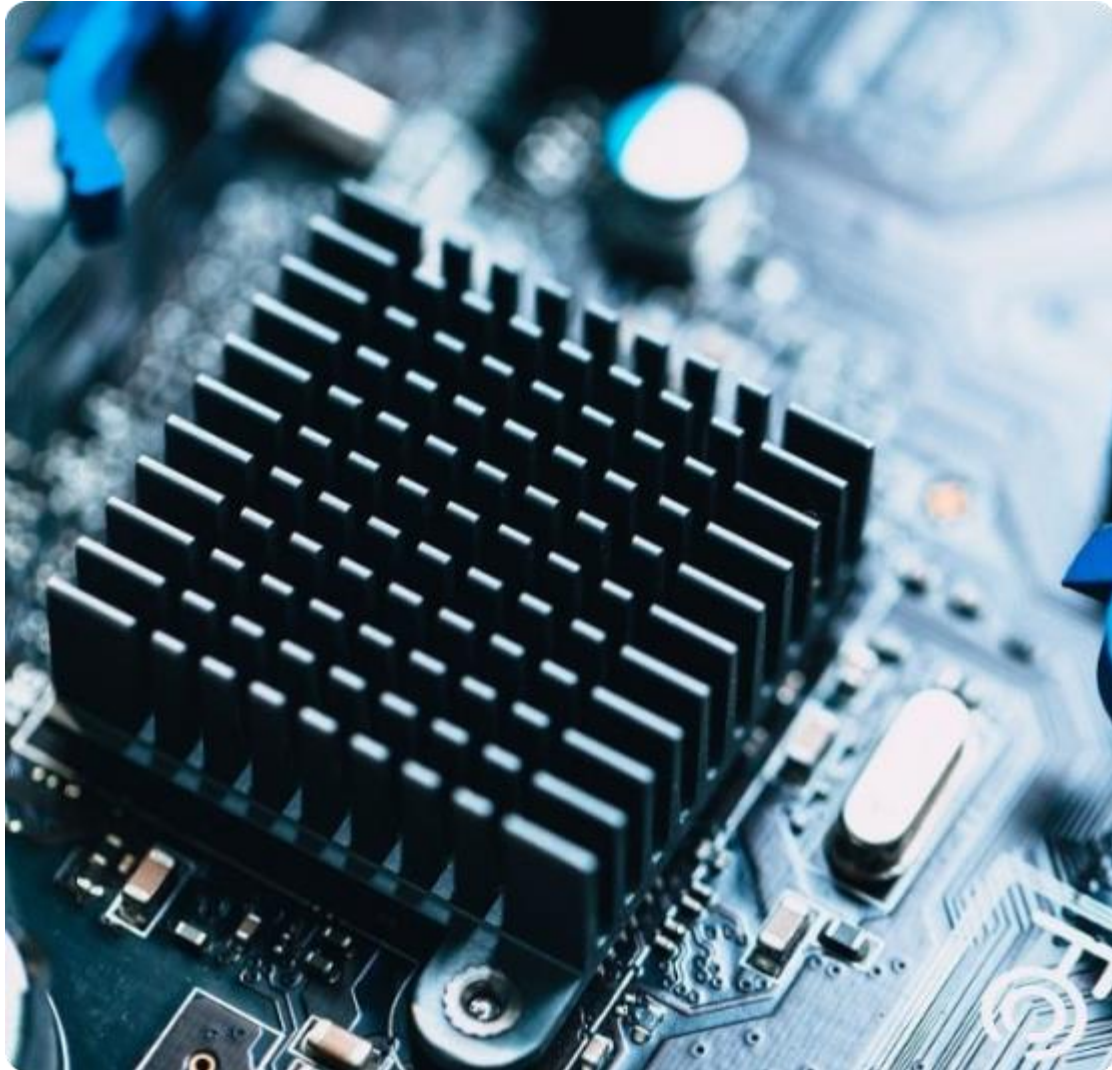


Charge Storage Memory

DRAM cells store individual bits as electrical charges inside microscopic capacitors.



Circuit Dynamics, Clock Speeds & Heat



RC Switching Delay

Interconnect resistance and capacitance set upper physical bounds on CPU clock frequencies.

Joule Heat Dissipation

Current flow produces thermal energy limiting power density in modern GPUs.

Multiphysics Co-Design

Thermal hotspots alter signal timing across multi-die microchip packaging.

Physical Layer & Network Signals

Fiber Optic Backbones

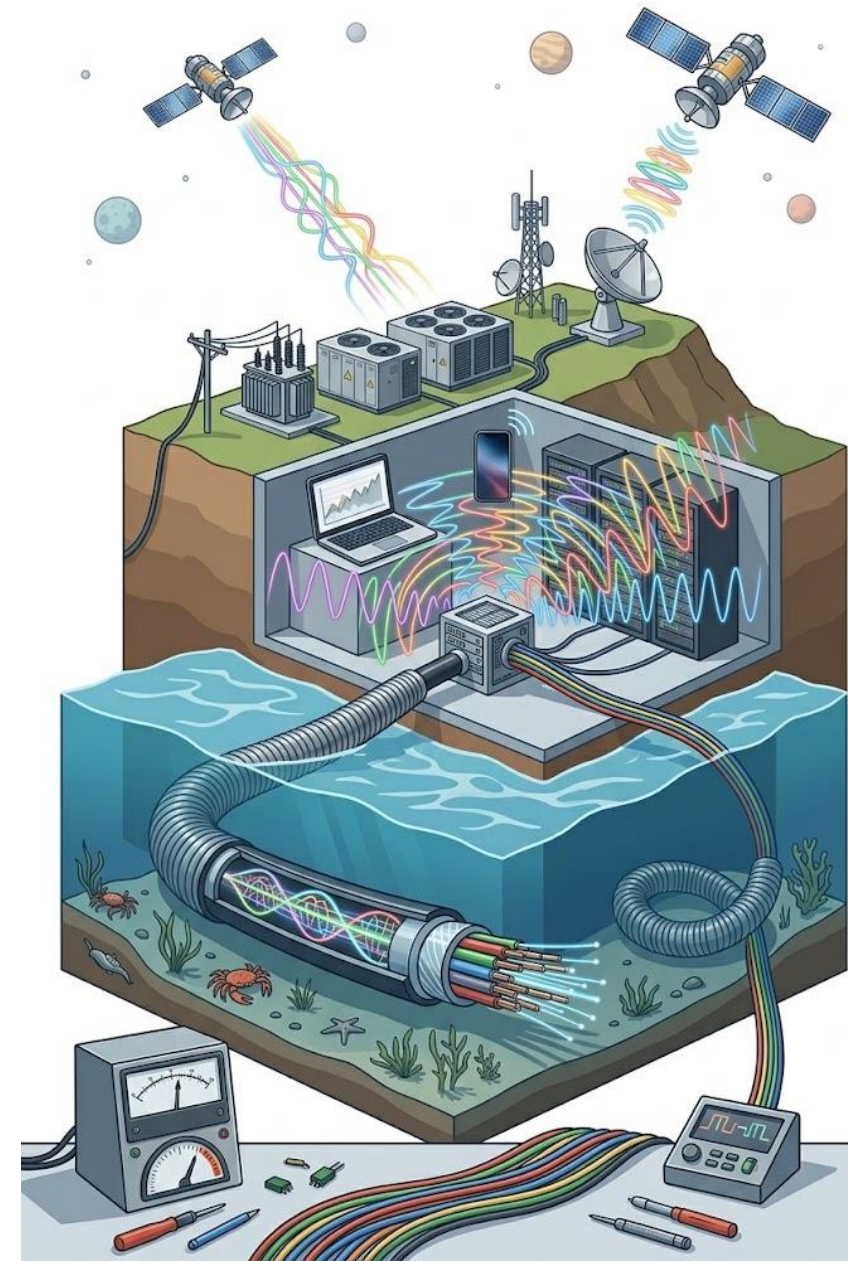
Total internal reflection enables high-speed global optical internet communications.

Electromagnetic Waves

Wi-Fi, 5G, and satellite protocols rely on radio wave modulation physics.

Signal Integrity

Electromagnetic interference and dispersion dictate real-world bus transfer rates.



Physics-Inspired AI & Machine Learning

✦ Diffusion Models

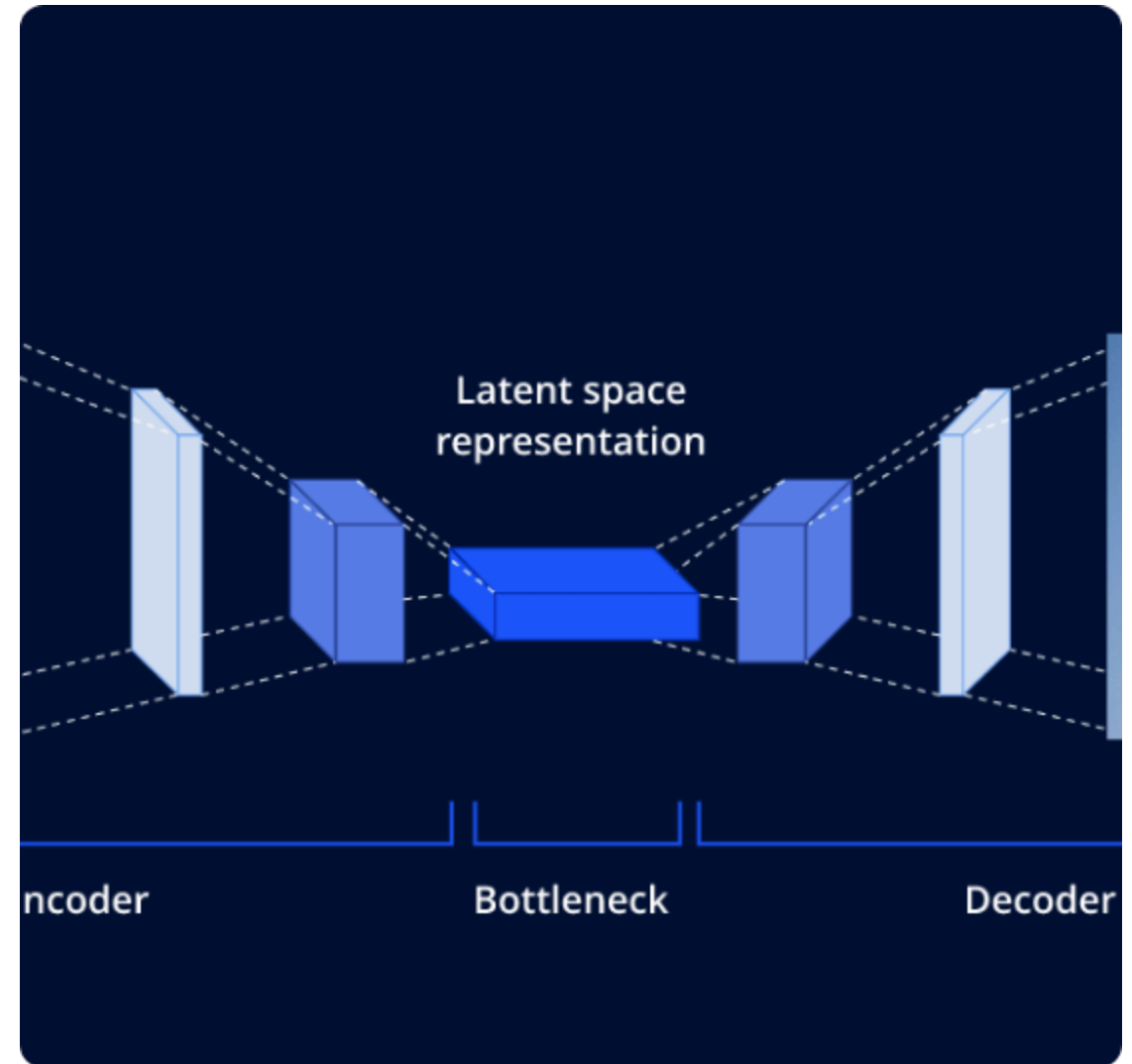
Generative AI models reverse thermodynamic gas diffusion to transform noise into images.

⚡ Energy-Based Networks

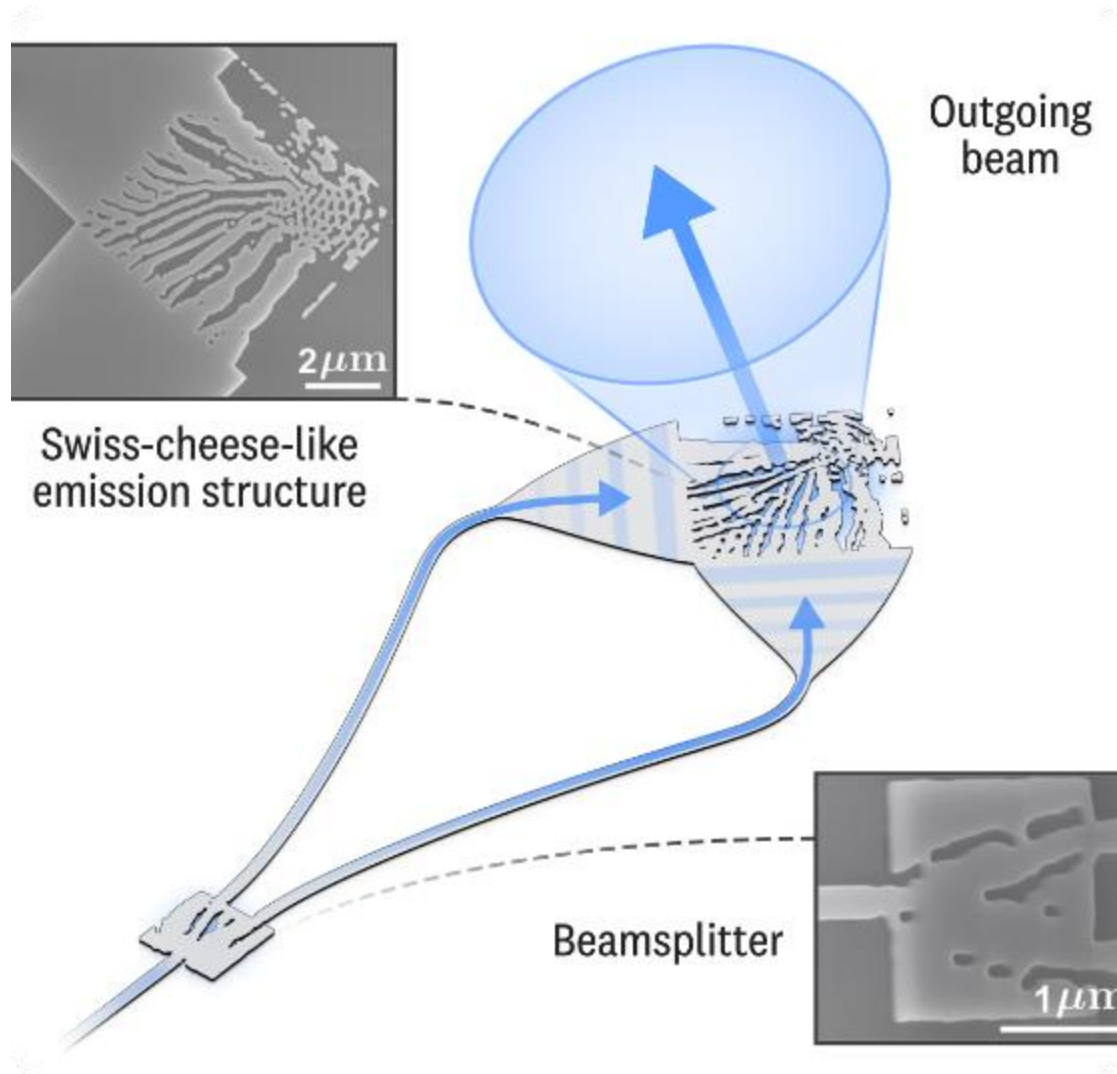
Hopfield nets adapt statistical mechanics Ising models for energy minimization.

🧠 Physics-Informed Neural Nets

PINNs embed conservation laws directly into loss functions to model physical systems.



Photonic Chips & Physical AI



💡 Photonic Computing

Light beams replace electrons to execute matrix math with ultra-low thermal dissipation.

⚙️ Coupled Oscillators

Self-organizing material networks compute pattern recognition directly in hardware.

🧠 Neuromorphic Memristors

Variable resistance materials store and process data simultaneously like biological synapses.

PILLAR 6: QUANTUM FRONTIERS

Quantum Limits & Quantum Computing

Quantum Tunneling

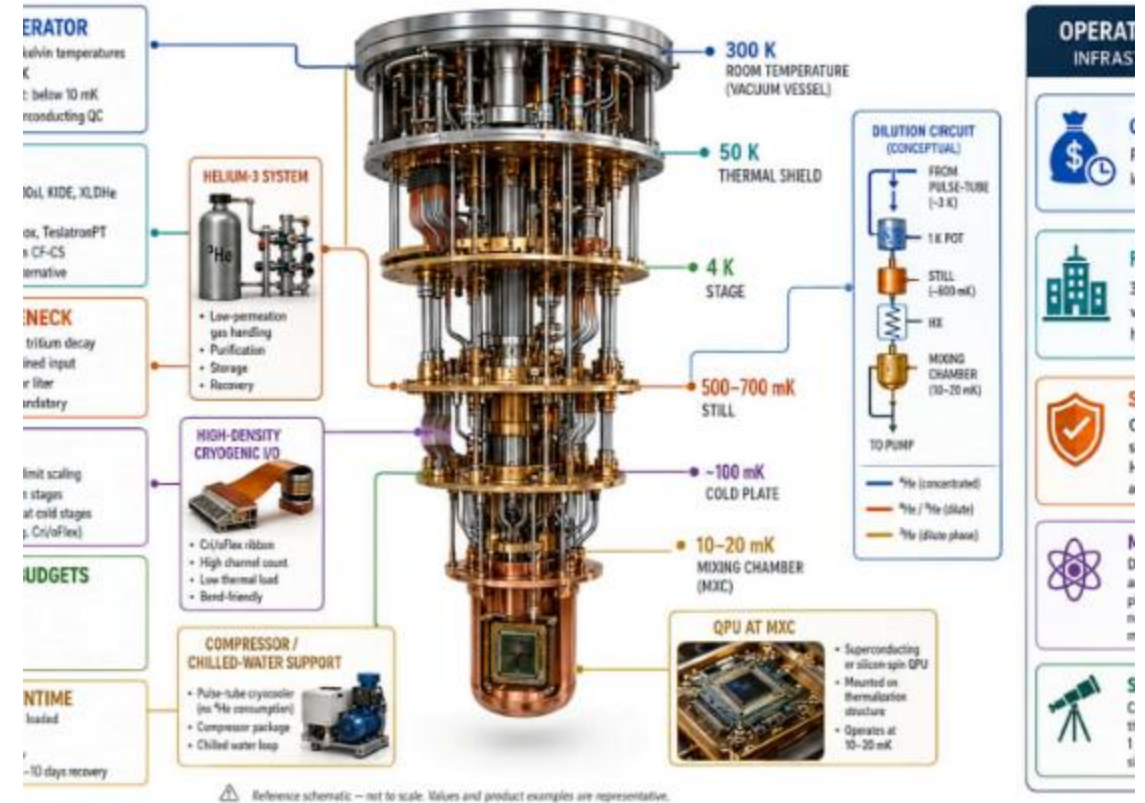
Electrons leak across sub-3nm transistor gates, forcing new physical architecture designs.

Superposition & Qubits

Quantum bits occupy combined basis states to perform exponential parallel calculations.

Entanglement Acceleration

Correlated quantum states enable revolutionary cryptographic and optimization algorithms.





Classical and Quantum Cryptography



Rail Fence cipher

Giving ciphertext

MEMATRHPRYETEFETEAT

Rail Fence cipher

Giving ciphertext

MEMATRHPRYETEFETEAT

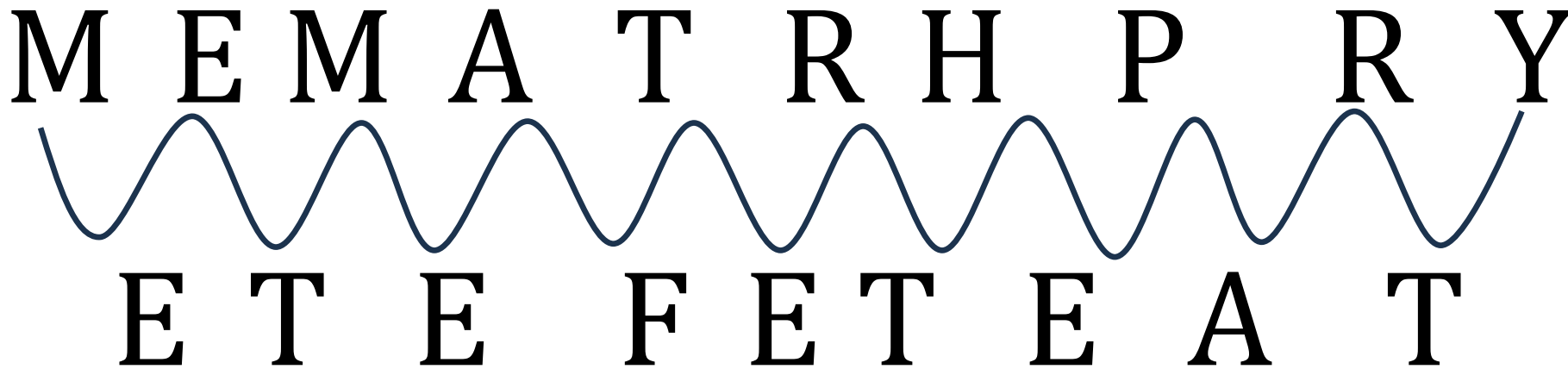
M E M A T R H P R Y

E T E F E T E A T

Rail Fence cipher

Giving ciphertext

MEMATRHPRYETEFETEAT



The diagram illustrates the Rail Fence cipher with two rows of letters. The top row contains the letters M, E, M, A, T, R, H, P, R, Y. The bottom row contains the letters E, T, E, F, E, T, E, A, T. A blue wavy line connects the letters in the top row to the letters in the bottom row, showing the interleaving pattern. The letters in the top row are connected to the letters in the bottom row by a series of peaks and valleys, representing the 'rails' of the cipher.

MEET ME AFTER THE PARTY

Row Transpose Cipher

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

Key: 4312567

Row Transpose Cipher

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

Key: 4312567

T	T	N	A	A	P	T	M	T	S	U	O	A	O	D	W	C	O	I	X	K	N	L	Y	P	E	T	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Row Transpose Cipher

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

Key: 4312567

	A	P	T	M	T	S	U	O	A	O	D	W	C	O	I	X	K	N	L	Y	P	E	T	Z
--	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

T
 T
 N
 A

Row Transpose Cipher

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

Key: 4312567

--	--	--	--	--	--	--

1	2	3	4	5	6	7
T	A	T	A	C	K	P
T	P	S	O	O	N	E
N	T	U	D	I	L	T
A	M	O	W	X	Y	Z

Row Transpose Cipher

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

Key: 4312567

--	--	--	--	--	--	--

4 3 1 2 5 6 7

A T T A C K P

O S T P O N E

D U N T I L T

W O A M X Y Z

Row Transpose Cipher

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

Key: 4312567

--	--	--	--	--	--	--

4 3 1 2 5 6 7

A T T A C K P

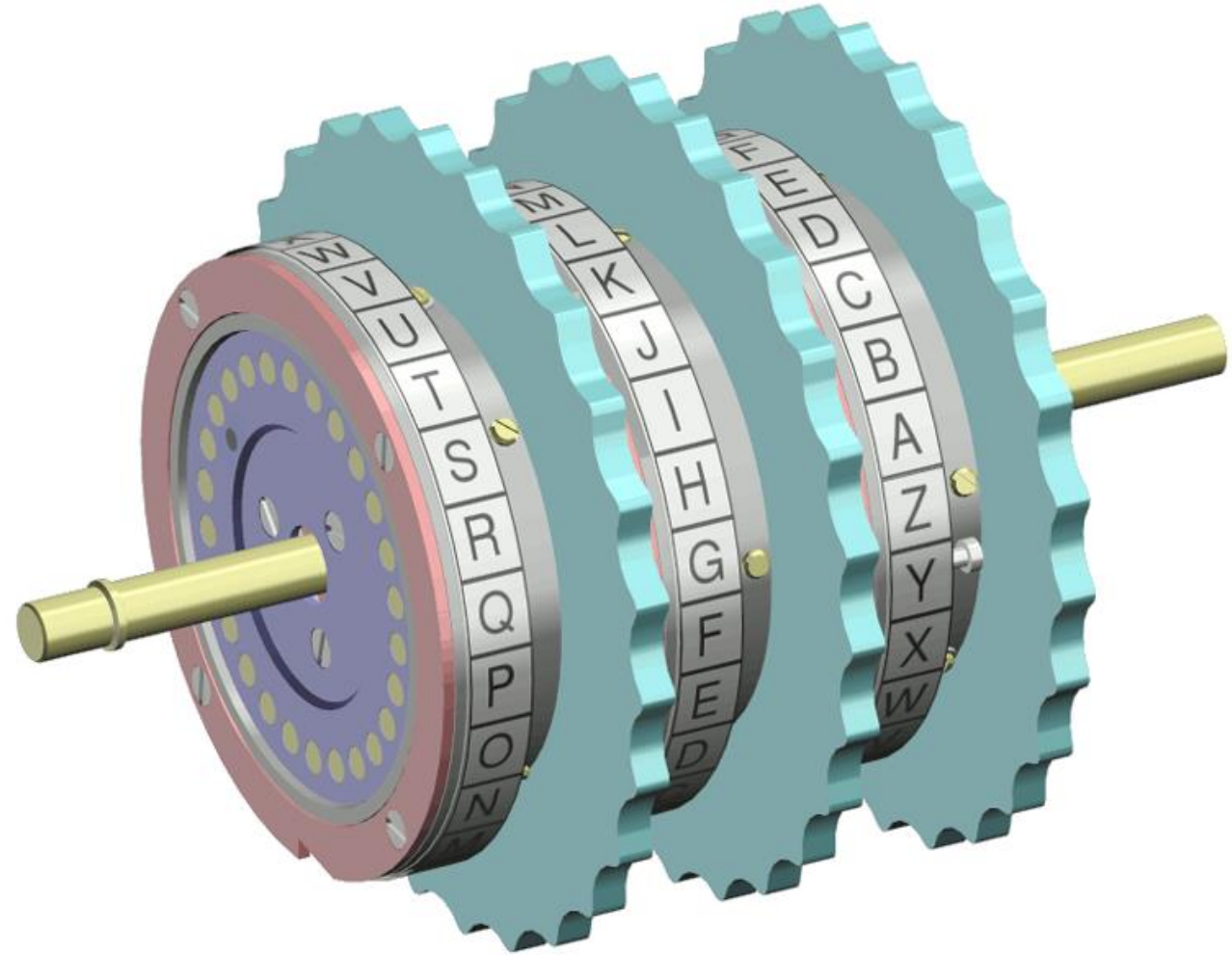
O S T P O N E

D U N T I L T

W O A M X Y Z

**Attack postponed Until
two AM XYZ**

The Rotors

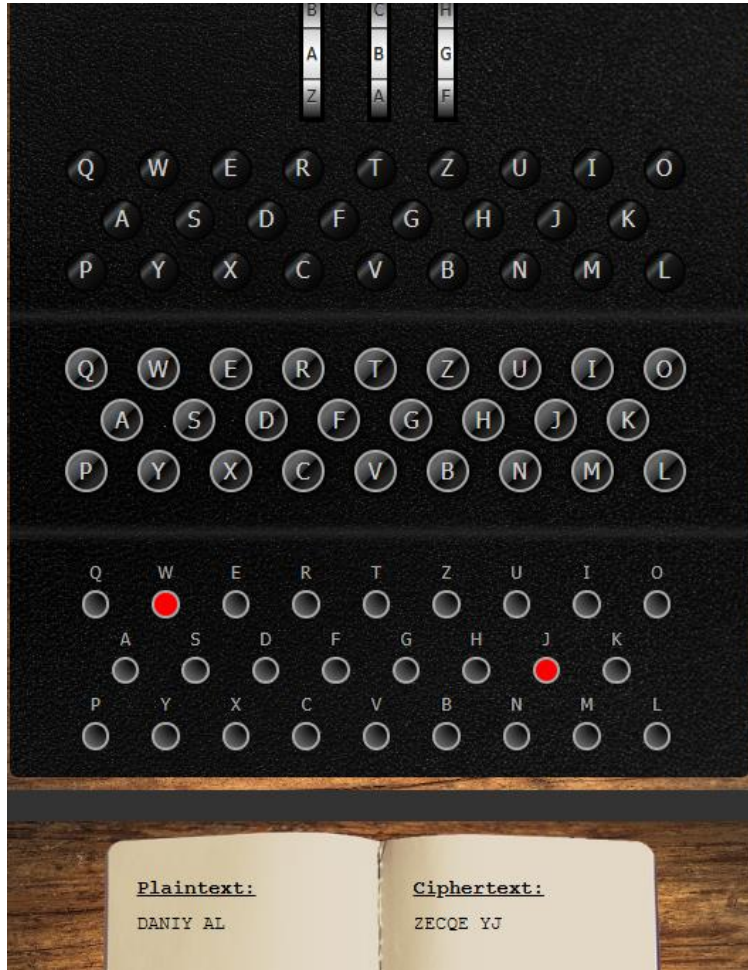


Rotor Cipher Machines

- Before modern ciphers, rotor machines were most common complex ciphers in use.
- Widely used in WW2.
- Used a series of rotating cylinders.
- Implemented a polyalphabetic substitution cipher of period K .
- With 3 cylinders, $K = 26^3 = 17,576$.
- With 5 cylinders, $K = 26^5 = 12 \times 10^6$.



Enigma decrypt by bombe



Simulator: <https://www.101computing.net/enigma-machine-emulator/>

Alan Turing



Quantum Cryptography

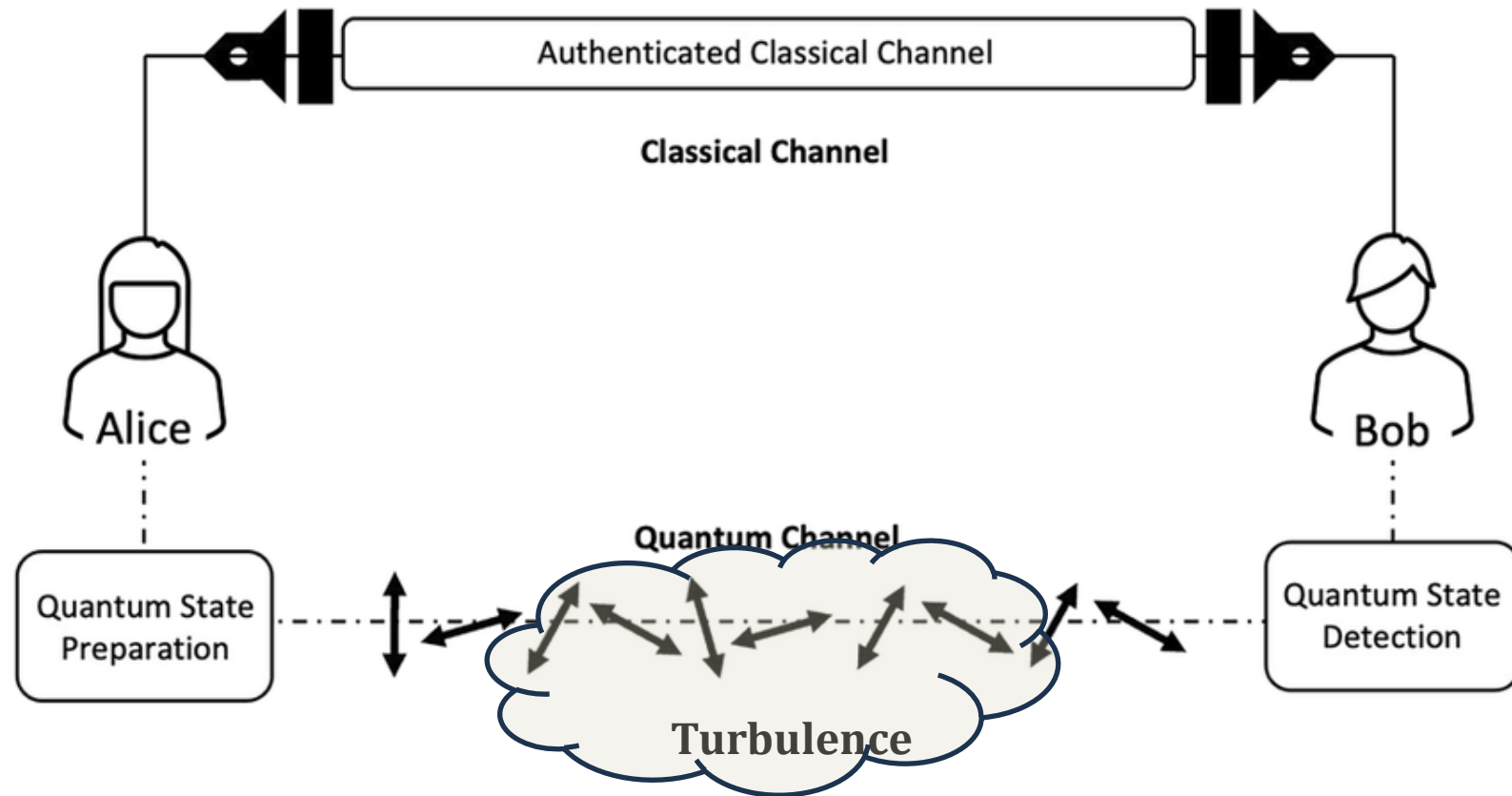
Quantum Key Distribution (QKD)

Quantum Key Distribution

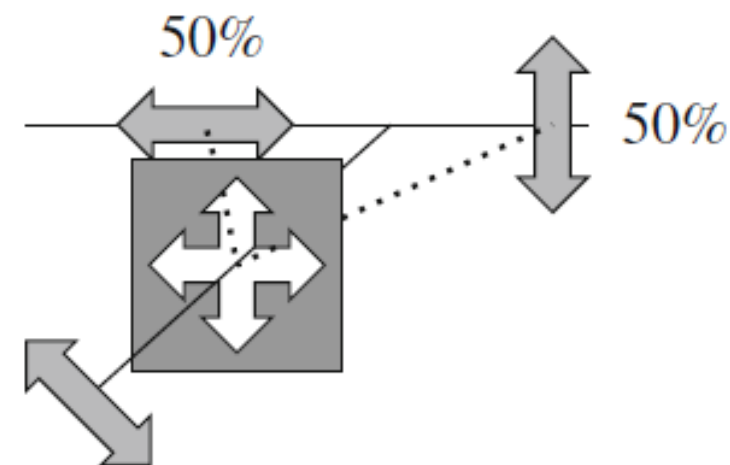
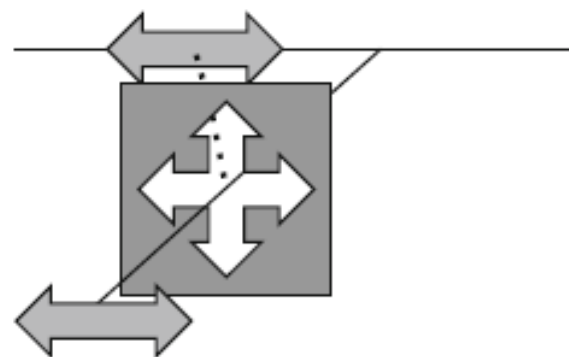
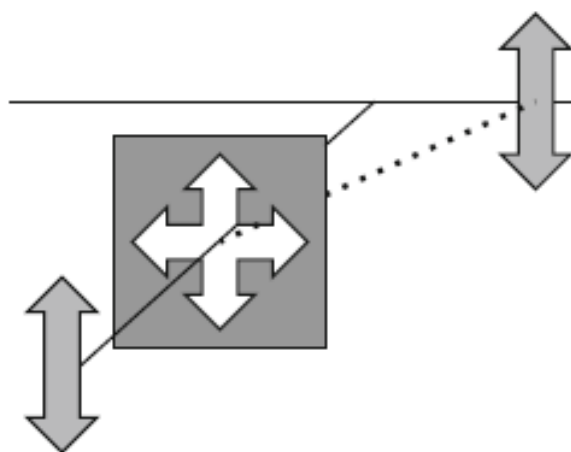
A secure communication method that uses quantum mechanics to generate and share secret cryptographic keys between two parties.

Types of Quantum Channel:

1. Fiber-optic channel
2. Free-space optical channel
3. Satellite-based channel



Bit value:	0	1
Rectilinear basis:		
Diagonal basis:		



Recommended text books

1. T. L. Floyd and D. L. Buchla, *Electronics Fundamentals: Circuits, Devices, and Applications*, 8th ed. Upper Saddle River, NJ: Pearson College Div, 2014.
2. R. L. Tokheim, *Digital Electronics: Principles and Applications*, 9th ed. New York, NY: McGraw-Hill Education, 2021.
3. D. Halliday, R. Resnick, and K. S. Krane, *Physics*, vol. 2, 5th ed. Hoboken, NJ: Wiley, 2010.
4. A. P. Malvino, D. J. Bates, and P. E. Hoppe, *Electronic Principles*, 8th ed. New York, NY: McGraw-Hill Education, 2015.
5. C. Bernhardt, *Quantum Computing for Everyone*. Cambridge, MA: MIT Press, 2019.

